

In the Name of God

Computational Fluid Dynamics II

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Chapter 5

Finite Volume Method (FVM)

فصل ۵

روش حجم محدود

روش حجم محدود

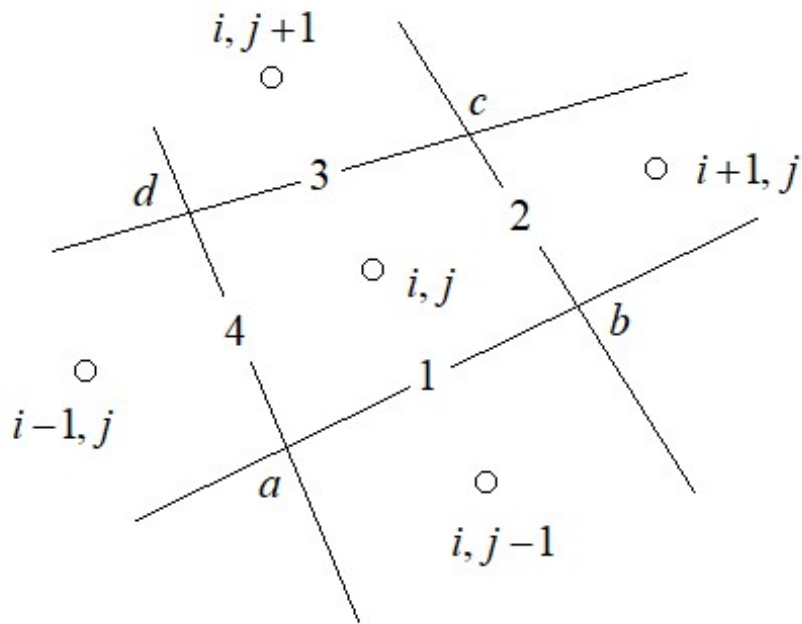
- ویژگیهای روش حجم محدود

- بیان معادلات حاکم برای المان کوچک محصور توسط یک سطح/حجم (گسسته‌سازی شکل انتگرالی)
- ارضاء قوانین بقا در فرم انتگرالی
- عدم نیاز به نگاشت (انتگرال‌گیری متغیرهای وابسته به‌طور مستقیم در دامنه فیزیکی)
- وابسته بودن دقت حل به نحوه تعیین شار روی وجوه المان
- قابل اعمال در شبکه‌های با‌سازمان و بی‌سازمان
- پیچیدگی افزایش دقت حل به ویژه در شبکه‌های بی‌سازمان

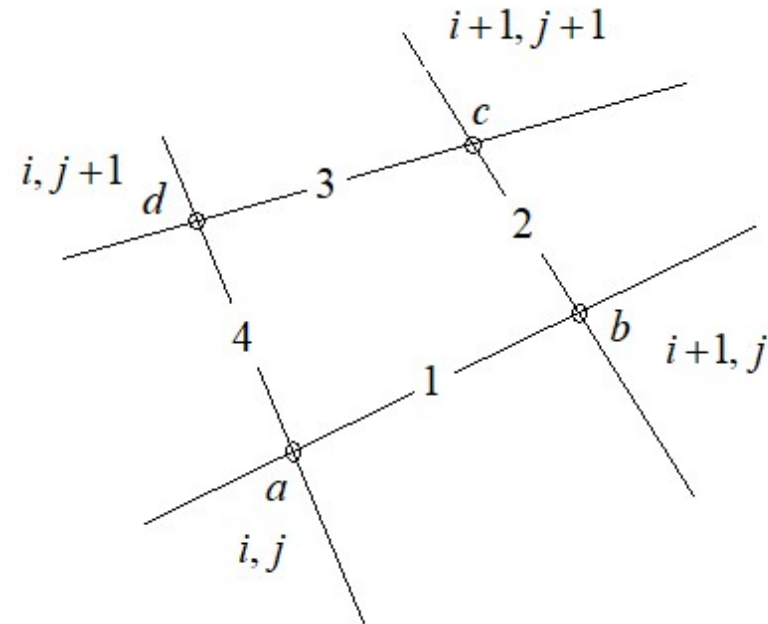
روش حجم محدود

- دسته‌بندی روش‌های حجم محدود

- تقسیم میدان به مجموعه‌ای از المان‌ها بدون هم‌پوشانی و اعمال قوانین بقا روی هر المان



Cell-centered method



Cell-vertex (nodal point) method

روش حجم محدود

- گسسته سازی معادلات اوایلر دوبعدی

2D unsteady compressible Euler eqs.:

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0 \quad \& \quad Q = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ E_t \end{pmatrix} \quad \& \quad F = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (E_t + p)u \end{pmatrix} \quad \& \quad G = \begin{pmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ (E_t + p)v \end{pmatrix}$$

$$\iint_A \frac{\partial Q}{\partial t} dA + \iint_A \left(\frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} \right) dA = 0$$

Green's theorem:

$$\oint_C F_1 dx + F_2 dy = \iint_A \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA \quad \rightarrow \quad \frac{\partial}{\partial t} \iint_A Q dA + \oint_C F dy - G dx = 0 \quad \&$$

$$\sum_{\text{cells}} \frac{\partial}{\partial t} \iint_A Q dA + \sum_{\text{cells}} \oint_C F dy - G dx = 0$$

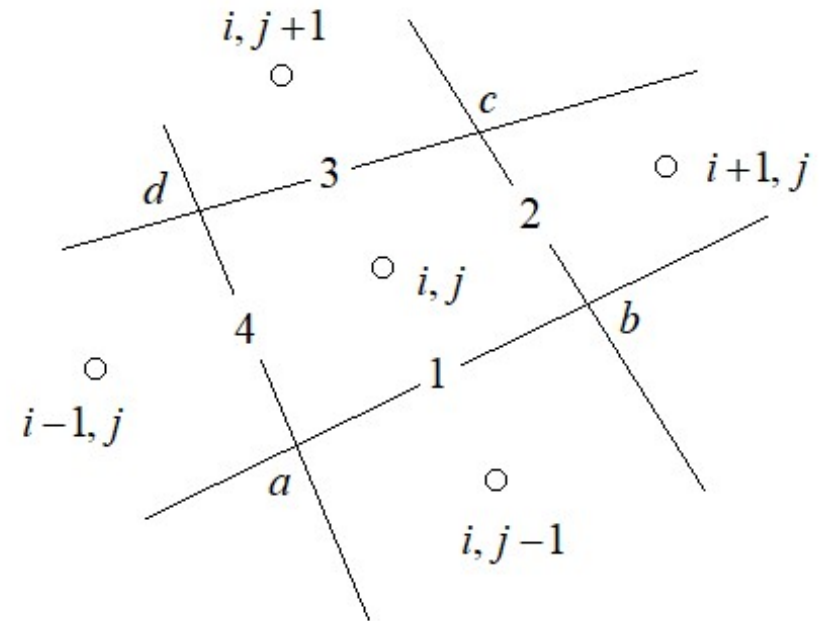
روش حجم محدود

$$\frac{\partial}{\partial t} \iint_A Q dA + \oint_C F dy - G dx = 0$$

• گسسته سازی معادلات اوایلر دوبعدی

Cell-centered scheme:

$$\iint_A \left(\frac{Q_{i,j}^{n+1} - Q_{i,j}^n}{\Delta t} \right) A_{i,j} + (F_1 \Delta y_1 + F_2 \Delta y_2 + F_3 \Delta y_3 + F_4 \Delta y_4) - (G_1 \Delta x_1 + G_2 \Delta x_2 + G_3 \Delta x_3 + G_4 \Delta x_4) = 0$$



$$\Delta x_1 = \Delta x_{ab} = x_b - x_a \quad \& \quad \Delta x_2 = \Delta x_{bc} = x_c - x_b \quad \dots$$

$$\text{Central differencing:} \quad F_1 = F_{ab} = \frac{F_{i,j}^* + F_{i,j-1}^*}{2} \quad \& \quad G_1 = G_{ab} = \frac{G_{i,j}^* + G_{i,j-1}^*}{2} \quad \dots$$

$$* = \begin{cases} n & \text{explicit scheme} \\ n+1 & \text{implicit scheme} \end{cases}$$

روش حجم محدود

- گسسته سازی معادلات اوایلر دوبعدی

$$\frac{\partial}{\partial t} \iint_A Q dA + \oint_C F dy - G dx = 0$$

$$\frac{d}{dt} (Q_{i,j} A_{i,j}) + H_{i,j} = 0$$

$$H_{i,j} = \text{net flux out of the cell} = \oint_C F dy - G dx = \begin{cases} \sum_{k=1}^4 (F_k \Delta y_k - G_k \Delta x_k) & \text{for quadrilateral cells} \\ \sum_{k=1}^3 (F_k \Delta y_k - G_k \Delta x_k) & \text{for triangle cells} \end{cases}$$

$$x\text{-momentum: } F_k = (\rho u^2 + p)_k \quad \& \quad G_k = (\rho uv)_k$$

$$H_{i,j} = \sum_{k=1}^4 \left[(\rho u^2 + p)_k \Delta y_k - (\rho uv)_k \Delta x_k \right] = \sum_{k=1}^4 \left[p_k \Delta y_k + (\rho u)_k (u \Delta y - v \Delta x)_k \right]$$

$$p_1 = p_{ab} = \frac{p_{i,j} + p_{i,j-1}}{2} \quad \& \quad (\rho u)_1 = (\rho u)_{ab} = \frac{(\rho u)_{i,j} + (\rho u)_{i,j-1}}{2} \quad \dots \rightarrow T.E = O[(\Delta x)^2, (\Delta y)^2]$$

روش حجم محدود

- گسسته سازی معادلات اوایلر دوبعدی

$$\frac{d}{dt}(Q_{i,j}A_{i,j}) + H_{i,j} = 0 \quad \& \quad H_{i,j} = \begin{pmatrix} \sum_{k=1}^4 [\rho_k (u\Delta y - v\Delta x)_k] \\ \sum_{k=1}^4 [p_k \Delta y_k + (\rho u)_k (u\Delta y - v\Delta x)_k] \\ \sum_{k=1}^4 [p_k \Delta x_k + (\rho v)_k (u\Delta y - v\Delta x)_k] \\ \sum_{k=1}^4 [(E_t + p)_k (u\Delta y - v\Delta x)_k] \end{pmatrix}$$

Slip wall boundary condition:

$$\frac{dy}{dx} = \frac{v}{u} \quad \rightarrow \quad (udy - vdx)_{k=1} = 0$$

روش حجم محدود

- انتگرال گیری زمانی به روش های رانگ-کوتا

- مناسب جهت حل مسائل ODE

- تبدیل PDE به Pseudo-PDE

Runge-Kutta schemes: $\frac{du}{dt} = R(u)$

$$\text{Wave equation: } \frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0 \quad \rightarrow \quad \frac{du}{dt} = R(u) \quad \& \quad R(u) = -c \frac{\partial u}{\partial x}$$

$$u^{(0)} = u^{(n)}$$

$$u^{(1)} = u^{(0)} + \alpha_1 \Delta t R(u^{(0)})$$

$$u^{(2)} = u^{(0)} + \alpha_2 \Delta t R(u^{(1)})$$

$\vdots$

$$u^{(k-1)} = u^{(0)} + \alpha_{k-1} \Delta t R(u^{(k-2)})$$

$$u^{(k)} = u^{(0)} + \alpha_k \Delta t R(u^{(k-1)})$$

$$u^{(n+1)} = u^{(k)}$$

روش حجم محدود

- انتگرال گیری زمانی به روش های رانگ-کوتا

- بررسی دقت روش رانگ-کوتا با حل مساله ODE خطی

$$\begin{cases} \frac{du}{dt} = R(u) = \lambda u \\ u(0) = u_0 \end{cases} \quad \& \quad \lambda = -a < 0 \quad \& \quad u(t)|_{\text{exact}} = u_0 e^{\lambda t} = u_0 e^{-at}$$

$$u^{(0)} = u^{(n)}$$

$$u^{(1)} = u^{(0)} + \alpha_1 \Delta t R(u^{(0)}) = u^{(n)} + \alpha_1 \Delta t \lambda u^{(n)} = (1 + \alpha_1 \lambda \Delta t) u^{(n)}$$

$$u^{(2)} = u^{(0)} + \alpha_2 \Delta t R(u^{(1)}) = u^{(n)} + \alpha_2 \Delta t \lambda (1 + \alpha_1 \lambda \Delta t) u^{(n)} = [1 + \alpha_2 \lambda \Delta t (1 + \alpha_1 \lambda \Delta t)] u^{(n)}$$

$$\begin{aligned} u^{(3)} &= u^{(0)} + \alpha_3 \Delta t R(u^{(2)}) = u^{(n)} + \alpha_3 \Delta t \lambda [1 + \alpha_2 \lambda \Delta t (1 + \alpha_1 \lambda \Delta t)] u^{(n)} \\ &= [1 + \alpha_3 \lambda \Delta t (1 + \alpha_2 \lambda \Delta t (1 + \alpha_1 \lambda \Delta t))] u^{(n)} \end{aligned}$$

$$\begin{aligned} u^{(4)} &= u^{(0)} + \alpha_4 \Delta t R(u^{(3)}) = u^{(n)} + \alpha_4 \Delta t \lambda [1 + \alpha_3 \lambda \Delta t (1 + \alpha_2 \lambda \Delta t (1 + \alpha_1 \lambda \Delta t))] u^{(n)} \\ &= [1 + \alpha_4 \lambda \Delta t (1 + \alpha_3 \lambda \Delta t (1 + \alpha_2 \lambda \Delta t (1 + \alpha_1 \lambda \Delta t)))] u^{(n)} \end{aligned}$$

$$u^{(n+1)} = u^{(4)} = [1 + \alpha_4 \lambda \Delta t + \alpha_4 \alpha_3 (\lambda \Delta t)^2 + \alpha_4 \alpha_3 \alpha_2 (\lambda \Delta t)^3 + \alpha_4 \alpha_3 \alpha_2 \alpha_1 (\lambda \Delta t)^4] u^{(n)}$$

روش حجم محدود

- انتگرال گیری زمانی به روش های رانگ-کوتا
- بررسی دقت روش رانگ-کوتا با حل مساله ODE خطی

$$u(t)|_{\text{exact}} = u_0 e^{\lambda t}$$

$$u^{(n+1)}|_{\text{exact}} = \left[1 + (\lambda \Delta t) + \frac{1}{2}(\lambda \Delta t)^2 + \frac{1}{6}(\lambda \Delta t)^3 + \frac{1}{24}(\lambda \Delta t)^4 + \dots \right] u^{(n)}$$

$$u^{(n+1)}|_{\text{num}} = \left[1 + \alpha_4 (\lambda \Delta t) + \alpha_4 \alpha_3 (\lambda \Delta t)^2 + \alpha_4 \alpha_3 \alpha_2 (\lambda \Delta t)^3 + \alpha_4 \alpha_3 \alpha_2 \alpha_1 (\lambda \Delta t)^4 \right] u^{(n)}$$

$$\begin{cases} \alpha_1 = \frac{1}{4}, \quad \alpha_2 = \frac{1}{3}, \quad \alpha_3 = \frac{1}{2}, \quad \alpha_4 = 1, & T.E = O(\Delta t)^4 & \text{4th-order R-K} \\ \alpha_1 = \frac{1}{2}, \quad \alpha_2 = \frac{1}{2}, \quad \alpha_3 = 1, \quad \alpha_4 = 1, & T.E = O(\Delta t)^2 & \text{2nd-order R-K} \end{cases}$$

روش حجم محدود

- انتگرال گیری زمانی به روش های رانگ-کوتا
- بررسی پایداری روش رانگ-کوتا با حل مساله ODE خطی

$$\begin{cases} \frac{du}{dt} = R(u) = \lambda u \\ u(0) = u_0 \end{cases}$$

Euler explicit scheme:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = R(u_i^n) = \lambda u_i^n \quad \rightarrow \quad u_i^{n+1} = u_i^n (1 + \lambda \Delta t)$$

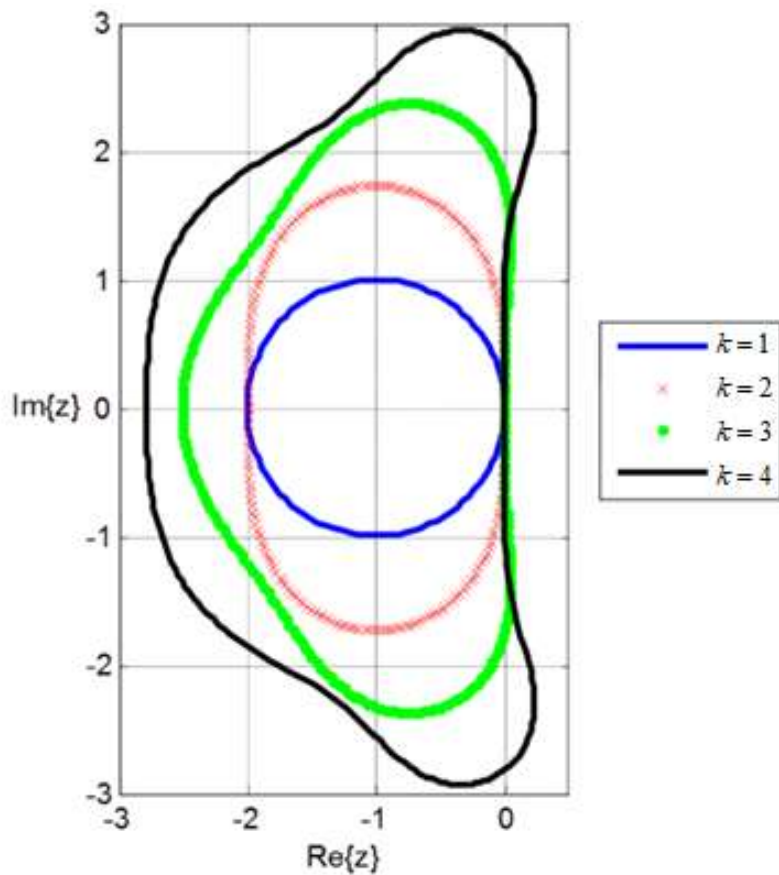
$$|g| = \text{amplification factor} = \left| \frac{u_i^{n+1}}{u_i^n} \right| = |1 + \lambda \Delta t| = |z| \leq 1 \quad \rightarrow \quad -1 \leq 1 + \lambda \Delta t \leq 1 \quad \rightarrow \quad 0 \leq \Delta t \leq \frac{-2}{\lambda} \quad \& \quad \lambda < 0$$

4th-order R-K :

$$u^{n+1} = \left[1 + (\lambda \Delta t) + \frac{1}{2}(\lambda \Delta t)^2 + \frac{1}{6}(\lambda \Delta t)^3 + \frac{1}{24}(\lambda \Delta t)^4 \right] u^n$$

$$|g| = \text{amplification factor} = \left| \frac{u_i^{n+1}}{u_i^n} \right| = \left| 1 + (\lambda \Delta t) + \frac{1}{2}(\lambda \Delta t)^2 + \frac{1}{6}(\lambda \Delta t)^3 + \frac{1}{24}(\lambda \Delta t)^4 \right| = |z| \leq 1$$

روش حجم محدود



- انتگرال گیری زمانی به روش های رانگ-کوتا
- بررسی پایداری روش رانگ-کوتا با حل مساله ODE خطی

- بررسی پایداری روش رانگ-کوتای مرتبه چهار برای مساله موج

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0 \quad \rightarrow \quad CFL = \frac{c \Delta t}{\Delta x} \leq 2\sqrt{2}$$

- بررسی پایداری روش رانگ-کوتای مرتبه چهار برای معادلات اوایلر یک بعدی

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} = 0 \quad \rightarrow \quad CFL = \frac{(u + a) \Delta t}{\Delta x} \leq 2\sqrt{2}$$

روش حجم محدود

• گسسته سازی معادلات اوایلر دوبعدی

Jameson's Runge-Kutta time-stepping method:

$$\frac{d}{dt}(Q_{i,j}A_{i,j}) + H_{i,j} - D_{i,j} = 0 \quad \& \quad D_{i,j} = \text{dissipation (flux) vector}$$

- روش جیمسون

$$\frac{dQ_{i,j}}{dt} = R(Q_{i,j}) \quad \& \quad R(Q_{i,j}) = -\frac{1}{A_{i,j}}(H_{i,j} - D_{i,j})$$

$$Q^{(0)} = Q^{(n)}$$

$$Q^{(1)} = Q^{(0)} + \alpha_1 \Delta t R(Q^{(0)})$$

$$Q^{(2)} = Q^{(0)} + \alpha_2 \Delta t R(Q^{(1)})$$

⋮

$$Q^{(k-1)} = Q^{(0)} + \alpha_{k-1} \Delta t R(Q^{(k-2)})$$

$$Q^{(k)} = Q^{(0)} + \alpha_k \Delta t R(Q^{(k-1)})$$

$$Q^{(n+1)} = Q^{(k)}$$

$$R(Q_{i,j}^{(k)}) = -\frac{1}{A_{i,j}}(H_{i,j}^{(k)} - D_{i,j}^{(0)})$$

روش حجم محدود

• گسسته سازی موج یک بعدی

- فرم عبارت میرایی در روش اختلاف محدود مرکزی

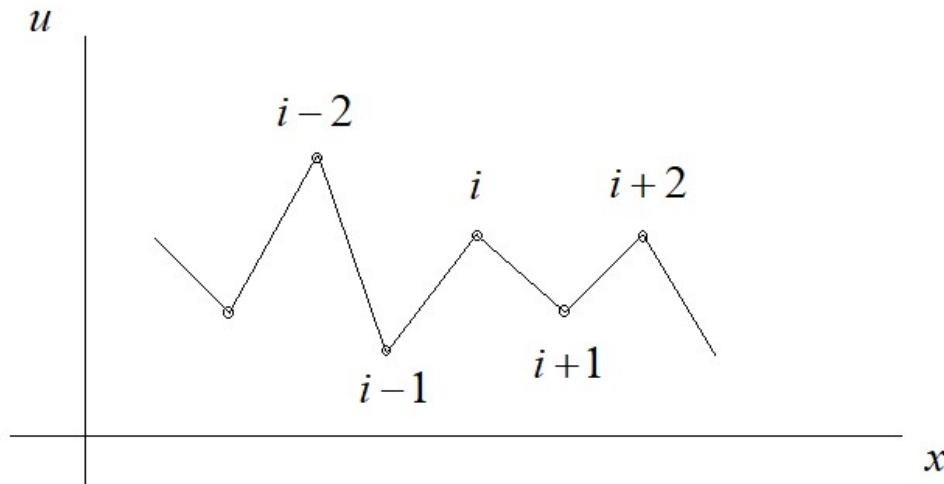
$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$$

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + c \frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x} = D_i^n \quad \& \quad D = \text{dissipation term}$$

$$u_i^{n+1} = u_i^n - c \frac{\Delta t}{2\Delta x} (u_{i+1}^n - u_{i-1}^n) + \Delta t D_i^n$$

$$\begin{aligned} D_i^n &= + \frac{(\Delta x)^2}{\Delta t} \frac{\partial}{\partial x} \left(\varepsilon^{(2)} \frac{\partial u}{\partial x} \right)_i^n - \frac{(\Delta x)^4}{\Delta t} \frac{\partial}{\partial x} \left(\varepsilon^{(4)} \frac{\partial^3 u}{\partial x^3} \right)_i^n \\ &= + \frac{(\Delta x)^2}{\Delta t} \frac{\varepsilon_{i+\frac{1}{2}}^{(2)} \left(\frac{\partial u}{\partial x} \right)_{i+\frac{1}{2}}^n - \varepsilon_{i-\frac{1}{2}}^{(2)} \left(\frac{\partial u}{\partial x} \right)_{i-\frac{1}{2}}^n}{\Delta x} - \frac{(\Delta x)^4}{\Delta t} \frac{\varepsilon_{i+\frac{1}{2}}^{(4)} \left(\frac{\partial^3 u}{\partial x^3} \right)_{i+\frac{1}{2}}^n - \varepsilon_{i-\frac{1}{2}}^{(4)} \left(\frac{\partial^3 u}{\partial x^3} \right)_{i-\frac{1}{2}}^n}{\Delta x} \\ &= + \frac{1}{\Delta t} \left\{ \varepsilon_{i+\frac{1}{2}}^{(2)} (u_{i+1} - u_i) - \varepsilon_{i-\frac{1}{2}}^{(2)} (u_i - u_{i-1}) \right\}^n - \frac{1}{\Delta t} \left\{ \varepsilon_{i+\frac{1}{2}}^{(4)} (u_{i+2} - 3u_{i+1} + 3u_i - u_{i-1}) - \varepsilon_{i-\frac{1}{2}}^{(4)} (u_{i+1} - 3u_i + 3u_{i-1} - u_{i-2}) \right\}^n \\ &= + \frac{1}{\Delta t} \varepsilon^{(2)} (u_{i+1} - 2u_i + u_{i-1})^n - \frac{1}{\Delta t} \varepsilon^{(4)} (u_{i+2} - 4u_{i+1} + 6u_i - 4u_{i-1} + u_{i-2})^n \quad \text{for } \varepsilon^{(2)} = \text{const.} \quad \& \quad \varepsilon^{(4)} = \text{const.} \end{aligned}$$

روش حجم محدود



• گسسته سازی موج یک بعدی

- فرم عبارت میرایی در روش اختلاف محدود مرکزی

$$u_i^{n+1} = u_i^n - c \frac{\Delta t}{2\Delta x} (u_{i+1}^n - u_{i-1}^n) + \Delta t D_i^n$$

$$D_i = +\frac{1}{\Delta t} \varepsilon^{(2)} (u_{i+1} - 2u_i + u_{i-1})^n - \frac{1}{\Delta t} \varepsilon^{(4)} (u_{i+2} - 4u_{i+1} + 6u_i - 4u_{i-1} + u_{i-2})^n \quad \text{for } \varepsilon^{(2)} = \text{const.} \quad \& \quad \varepsilon^{(4)} = \text{const.}$$

$$u_i^{n+1} = u_i^n - c \frac{\Delta t}{2\Delta x} (u_{i+1}^n - u_{i-1}^n) + \varepsilon^{(2)} (u_{i+1} - 2u_i + u_{i-1})^n - \varepsilon^{(4)} (u_{i+2} - 4u_{i+1} + 6u_i - 4u_{i-1} + u_{i-2})^n$$

$$= u_i^n - c \frac{\Delta t}{2\Delta x} (u_{i+1}^n - u_{i-1}^n)$$

$$+ \varepsilon^{(2)} \left\{ (u_{i+1} - u_i) + (u_{i-1} - u_i) \right\}^n$$

$$- \varepsilon^{(4)} \left\{ (u_{i+2} - u_{i+1}) + (u_{i-2} - u_{i-1}) + 3(u_i - u_{i+1}) + 3(u_i - u_{i-1}) \right\}^n$$

For stability: $\varepsilon^{(2)} > 0$ & $\varepsilon^{(4)} > 0$

روش حجم محدود

$$\frac{dQ_{i,j}}{dt} = R(Q_{i,j}) \quad \& \quad R(Q_{i,j}) = -\frac{1}{A_{i,j}}(H_{i,j} - D_{i,j})$$

$$D_{i,j} = (D_{i,j})_x + (D_{i,j})_y = d_{i+\frac{1}{2},j} - d_{i-\frac{1}{2},j} + d_{i,j+\frac{1}{2}} - d_{i,j-\frac{1}{2}}$$

• گسسته سازی معادلات اوایلر دوبعدی

$$d_{i+\frac{1}{2},j} = \frac{A_{i+\frac{1}{2},j}}{\Delta t} \left(\varepsilon_{i+\frac{1}{2},j}^{(2)} \delta Q_{i+\frac{1}{2},j} - \varepsilon_{i+\frac{1}{2},j}^{(4)} \delta Q_{i+\frac{1}{2},j}^3 \right)$$

- عبارت میرایی (لزجت مصنوعی)

$$= \frac{A_{i+\frac{1}{2},j}}{\Delta t} \left(\varepsilon_{i+\frac{1}{2},j}^{(2)} (Q_{i+1,j} - Q_{i,j}) - \varepsilon_{i+\frac{1}{2},j}^{(4)} (Q_{i+2,j} - 3Q_{i+1,j} + 3Q_{i,j} - Q_{i-1,j}) \right)$$

$$d_{i-\frac{1}{2},j} = \frac{A_{i-\frac{1}{2},j}}{\Delta t} \left(\varepsilon_{i-\frac{1}{2},j}^{(2)} \delta Q_{i-\frac{1}{2},j} - \varepsilon_{i-\frac{1}{2},j}^{(4)} \delta Q_{i-\frac{1}{2},j}^3 \right)$$

$$= \frac{A_{i-\frac{1}{2},j}}{\Delta t} \left(\varepsilon_{i-\frac{1}{2},j}^{(2)} (Q_{i,j} - Q_{i-1,j}) - \varepsilon_{i-\frac{1}{2},j}^{(4)} (Q_{i+1,j} - 3Q_{i,j} + 3Q_{i-1,j} - Q_{i-2,j}) \right)$$

$$\varepsilon_{i\pm\frac{1}{2},j}^{(2)} = k^{(2)} \max(\nu_{i\pm 1,j}, \nu_{i,j}) \quad \& \quad \varepsilon_{i\pm\frac{1}{2},j}^{(4)} = k^{(4)} \max\left(0, k^{(4)} - \varepsilon_{i\pm\frac{1}{2},j}^{(2)}\right) \quad \& \quad k^{(2)} = \frac{1}{4} \quad \& \quad k^{(4)} = \frac{1}{256}$$

$$\nu_{i,j} = \text{shock detecting sensor} = \frac{|p_{i+1,j} - 2p_{i,j} + p_{i-1,j}|}{p_{i+1,j} + 2p_{i,j} + p_{i-1,j}}$$

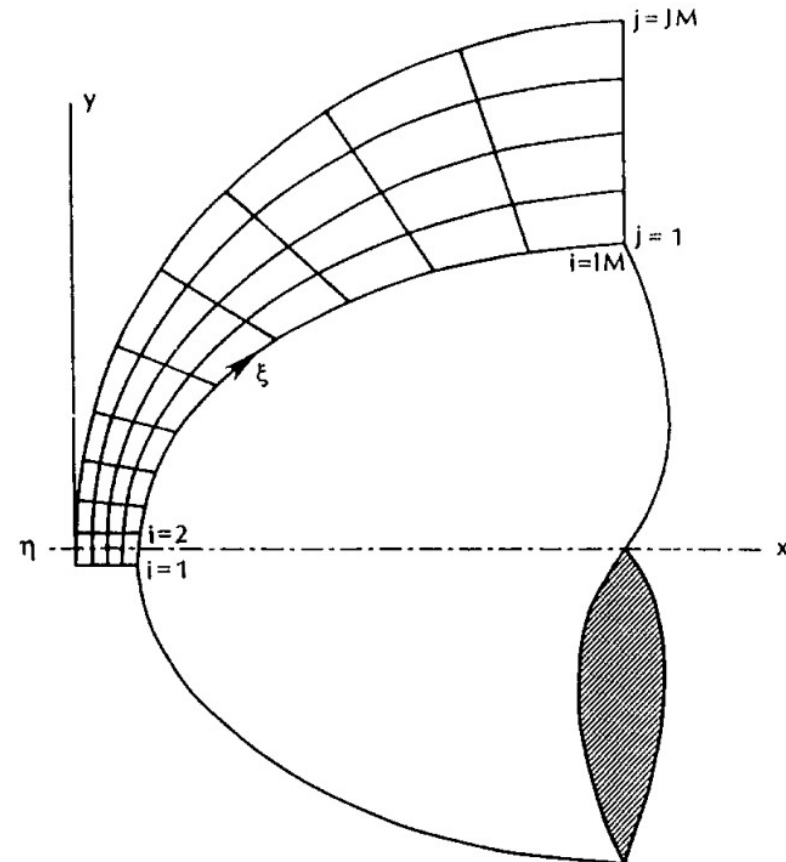
روش حجم محدود

- گسسته‌سازی معادلات اوایلر دوبعدی

- شرایط مرزی

Symmetry B.C :

$$\left\{ \begin{array}{l} \frac{\partial p}{\partial n} = 0 \\ \frac{\partial \rho}{\partial n} = 0 \\ \frac{\partial u}{\partial n} = 0 \\ v = 0 \end{array} \right. \rightarrow \left\{ \begin{array}{l} p_{i,1} = p_g \\ \rho_{i,1} = \rho_g \\ u_{i,1} = u_g \\ v_{i,1} = -v_g \end{array} \right.$$



روش حجم محدود

- گسسته سازی معادلات اوایلر دوبعدی

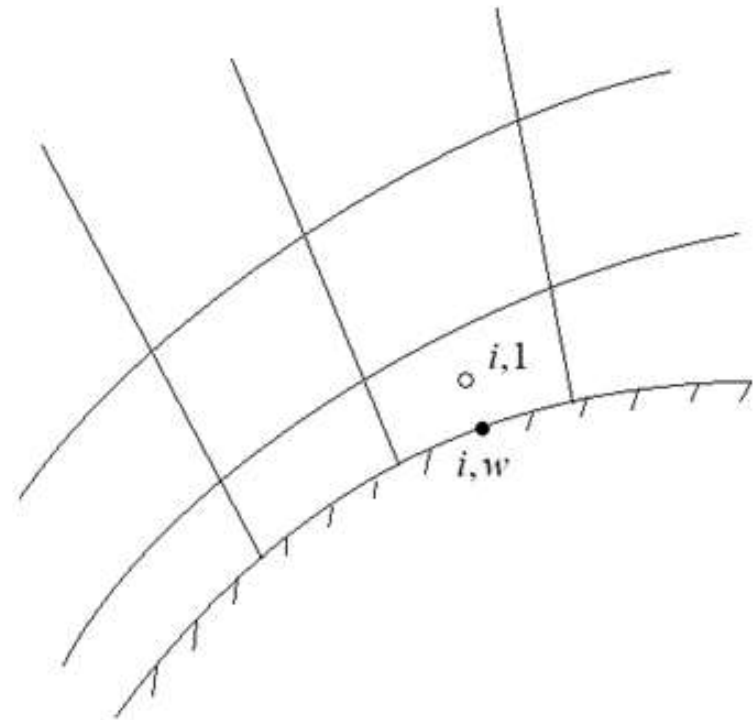
- شرایط مرزی

Slip wall B.C :

$$\frac{dy}{dx} = \frac{v}{u} \rightarrow (udy - vdx)_{k=1} = 0$$

$$\frac{\partial p}{\partial n} = \rho \frac{V_t^2}{R}$$

$$R \gg 1 \rightarrow \frac{\partial p}{\partial n} \approx 0 \rightarrow p_{i,1} = p_w$$



روش حجم محدود

• گسسته‌سازی معادلات اوایلر دوبعدی

- شرایط مرزی

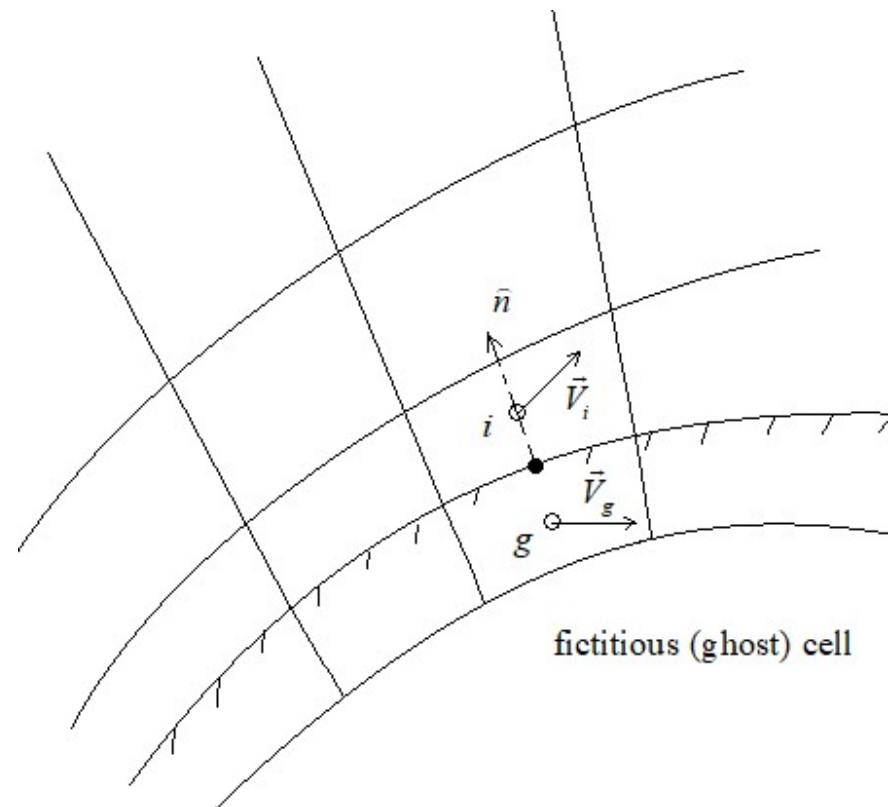
Wall B.C:

$$\begin{cases} V_{i,n} = \vec{V}_i \cdot \hat{n} = u_i n_x + v_i n_y \\ V_{g,n} = \vec{V}_g \cdot \hat{n} = u_g n_x + v_g n_y \end{cases}$$

$$\begin{cases} V_{i,n} + V_{g,n} = 0 \\ |V_i| = |V_g| \end{cases} \rightarrow \begin{cases} u_g n_x + v_g n_y = -(u_i n_x + v_i n_y) \\ u_g^2 + v_g^2 = u_i^2 + v_i^2 \end{cases}$$

$$\begin{pmatrix} u_g \\ v_g \end{pmatrix} = \begin{pmatrix} 1 - 2n_x^2 & -2n_x n_y \\ -2n_x n_y & 1 - 2n_y^2 \end{pmatrix} \begin{pmatrix} u_i \\ v_i \end{pmatrix} \quad \text{Inviscid (slip) wall}$$

$$\begin{pmatrix} u_g \\ v_g \end{pmatrix} = \begin{pmatrix} -u_i \\ -v_i \end{pmatrix} \quad \text{Viscous (noslip) wall}$$



روش حجم محدود

- گسسته‌سازی معادلات اوایلر دوبعدی

Characteristic B.Cs :

- شرایط مرزی

Subsonic inflow:

$$\begin{cases} R^+ = U_{n\infty} + \frac{2a_\infty}{\gamma-1} = U_n + \frac{2a}{\gamma-1} \\ R^- = U_{ni} - \frac{2a_i}{\gamma-1} = U_n - \frac{2a}{\gamma-1} \end{cases} \quad \& \quad \begin{cases} U_{n\infty} = \vec{V}_\infty \cdot \hat{n} = u_\infty n_x + v_\infty n_y \\ U_n = \vec{V} \cdot \hat{n} = un_x + vn_y \end{cases}$$

$$\begin{cases} U_n = \frac{1}{2}(R^+ + R^-) \\ a = \frac{\gamma-1}{4}(R^+ - R^-) \end{cases} \quad \& \quad \vec{V} = \vec{V}_\infty + (\vec{V} \cdot \hat{n} - \vec{V}_\infty \cdot \hat{n})\hat{n} \quad \text{or} \quad \begin{cases} u = u_\infty + n_x(U_n - U_{n\infty}) \\ v = v_\infty + n_y(U_n - U_{n\infty}) \end{cases}$$

$$\begin{cases} \rho = \frac{a^2}{\gamma s} \\ p = \frac{\rho a^2}{\gamma} \end{cases} \quad \& \quad s = \frac{p}{\rho^\gamma} = s_\infty = \frac{p_\infty}{\rho_\infty^\gamma} \quad \& \quad a^2 = \gamma \frac{p}{\rho}$$

روش حجم محدود

- معادلات ناویر-استوکس دوبعدی

2D unsteady compressible Navier-Stokes eqs.:

$$\frac{\partial Q}{\partial t} + \frac{\partial F_i}{\partial x} + \frac{\partial G_i}{\partial y} = \frac{1}{\text{Re}} \left(\frac{\partial F_v}{\partial x} + \frac{\partial G_v}{\partial y} \right)$$

$$Q = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ E_t \end{pmatrix}, F_i = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (E_t + p)u \end{pmatrix}, G_i = \begin{pmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ (E_t + p)v \end{pmatrix}, F_v = \begin{pmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u\tau_{xx} + v\tau_{xy} + \frac{\gamma}{\text{Pr}} k \frac{\partial e}{\partial x} \end{pmatrix}, G_v = \begin{pmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ u\tau_{xy} + v\tau_{yy} + \frac{\gamma}{\text{Pr}} k \frac{\partial e}{\partial y} \end{pmatrix}$$

$$E_t = \rho \left(e + \frac{1}{2} V^2 \right) = \frac{p}{\gamma - 1} + \frac{1}{2} \rho V^2 \quad \& \quad p = \rho T \quad \& \quad k = \mu$$

$$\text{Re} = \frac{\text{Re}_\infty}{\sqrt{\gamma}} \frac{1}{M_\infty} \quad \& \quad \text{Re}_\infty = \frac{\rho_\infty u_\infty L}{\mu_\infty}$$

روش حجم محدود

Cell-centered FVM:

• گسسته‌سازی معادلات ناویر-استوکس دوبعدی

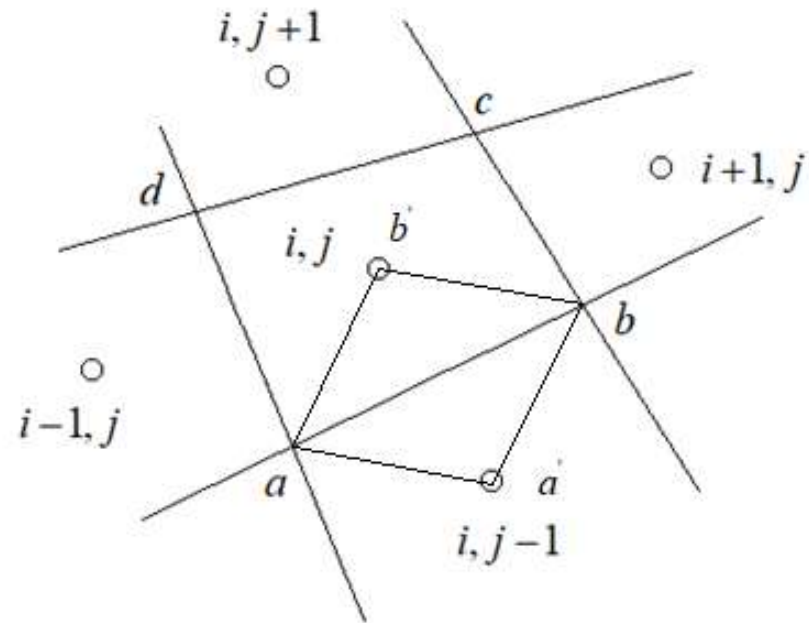
$$\frac{\partial Q}{\partial t} + \frac{\partial F_i}{\partial x} + \frac{\partial G_i}{\partial y} = \frac{1}{\text{Re}} \left(\frac{\partial F_v}{\partial x} + \frac{\partial G_v}{\partial y} \right)$$

$$\frac{\partial}{\partial t} \iint_A Q dA + \underbrace{\oint_C F_i dy - G_i dx}_{(H_i)_{i,j} = \sum_{k=1}^4 ((F_i)_k \Delta y_k - (G_i)_k \Delta x_k)} - \underbrace{\frac{1}{\text{Re}} \oint_C F_v dy - G_v dx}_{(H_v)_{i,j} = \frac{1}{\text{Re}} \sum_{k=1}^4 ((F_v)_k \Delta y_k - (G_v)_k \Delta x_k)} = 0$$

$$\frac{d}{dt} (Q_{i,j} A_{i,j}) + (H_i)_{i,j} - (H_v)_{i,j} - D_{i,j} = 0$$

$$\frac{dQ_{i,j}}{dt} = R(Q_{i,j}) \quad \& \quad R(Q_{i,j}) = -\frac{1}{A_{i,j}} \left\{ (H_i)_{i,j} - (H_v)_{i,j} - D_{i,j} \right\}$$

Central differencing: $(F_i)_1 = (F_i)_{ab} = \frac{(F_i)_{i,j} + (F_i)_{i,j-1}}{2} \quad \& \quad (G_i)_1 = (G_i)_{ab} = \frac{(G_i)_{i,j} + (G_i)_{i,j-1}}{2} \quad \dots$



روش حجم محدود

- گسسته‌سازی معادلات ناویر-استوکس دوبعدی

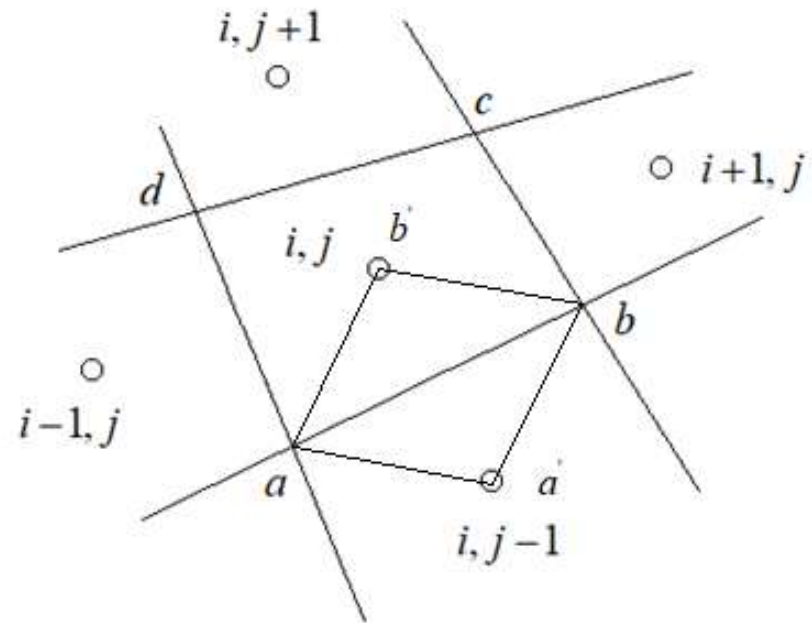
Cell-centered FVM:

$$(F_v)_1 = (F_v)_{ab} = ? \quad \& \quad (G_v)_1 = (G_v)_{ab} = ?$$

$$\left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}, \dots \right) = ?$$

$$\begin{aligned} \left(\frac{\partial u}{\partial x} \right)_{ab} &= \frac{1}{A_{aa'bb'}} \left[\int_{aa'bb'} \left(\frac{\partial u}{\partial x} \right) dx dy \right] \\ &= \frac{1}{2A_{aa'bb'}} \left[(u_a + u_{a'}) \Delta y_{aa'} + (u_{a'} + u_b) \Delta y_{a'b} + (u_b + u_{b'}) \Delta y_{bb'} + (u_{b'} + u_a) \Delta y_{b'a} \right] \end{aligned}$$

$$\begin{cases} \Delta x_{aa'} = x_{a'} - x_a, \dots \\ \Delta y_{aa'} = y_{a'} - y_a, \dots \end{cases} \quad \& \quad u_a = \frac{\sum_{m=1}^4 \frac{u_m}{L_{ma}}}{\sum_{m=1}^4 \frac{1}{L_{ma}}} \quad \& \quad L_{ma} = \sqrt{(x_m - x_a)^2 + (y_m - y_a)^2}$$



روش حجم محدود

- گسسته سازی معادلات ناویر-استوکس دوبعدی

Wall B.Cs :

$$\vec{V} = \vec{V}_{wall} \quad \text{at solid wall}$$

- شرایط مرزی

$$T = T_{wall} \quad \text{isothermal wall}$$

$$q_{wall} = -k \frac{\partial T}{\partial n} \quad \text{specified heat flux on wall}$$

$$q_{wall} = -k \frac{\partial T}{\partial n} = 0 \quad \text{adiabatic wall}$$

$$\frac{\partial p}{\partial n} = \frac{1}{Re} (\vec{\nabla} \cdot \underline{\underline{\tau}})_n \quad \rightarrow \quad \frac{\partial p}{\partial n} \approx 0 \quad \text{for} \quad Re \gg 1$$

Outflow B.C:

$$\frac{\partial^2 Q}{\partial x^2} \approx 0 \quad \text{extrapolation at outflow boundary}$$

روش حجم محدود

$$\frac{\partial u}{\partial t} + \frac{\partial F}{\partial x} = 0 \quad \& \quad F = \frac{u^2}{2} \quad \text{Traffic eq.}$$

$$\iint_A \left(\frac{\partial u}{\partial t} + \frac{\partial F}{\partial x} \right) dA = 0 \quad \& \quad dA = dx dt$$

Green's theorem:

$$\oint_C u dx - F dt = 0$$

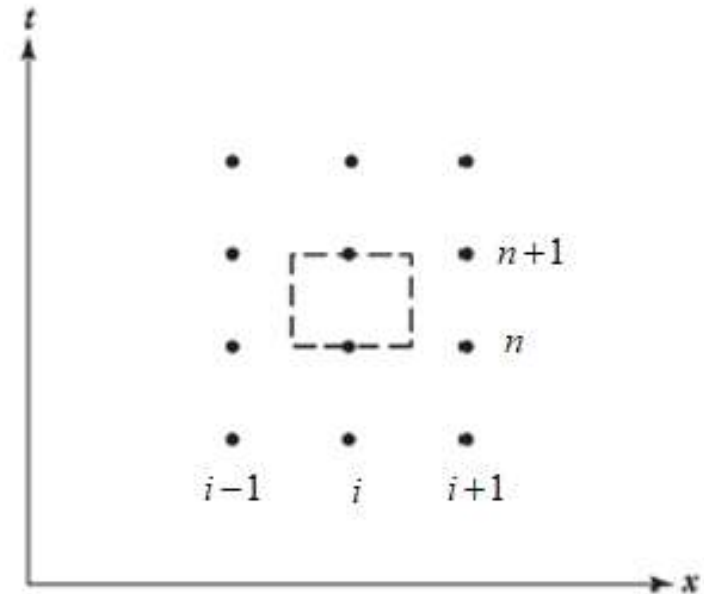
$$u_i^n \Delta x - u_i^{n+1} \Delta x - \left(f_{i+\frac{1}{2}}^{n+\frac{1}{2}} \Delta t - f_{i-\frac{1}{2}}^{n+\frac{1}{2}} \Delta t \right) = 0$$

$$u_i = \frac{1}{\Delta x} \int_{x-\frac{\Delta x}{2}}^{x+\frac{\Delta x}{2}} u(x,t) dx \quad \& \quad f^{n+\frac{1}{2}} = \frac{1}{\Delta t} \int_{\Delta t}^{t+\Delta t} F(x,t) dt$$

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + \frac{f_{i+\frac{1}{2}}^{n+\frac{1}{2}} - f_{i-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} = 0 \quad \text{for explicit method} \rightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} + \frac{f_{i+\frac{1}{2}}^n - f_{i-\frac{1}{2}}^n}{\Delta x} = 0$$

$$f_{i\pm\frac{1}{2}}^n = \text{numerical flux} = f(u_i, u_{i\pm 1}) \quad \& \quad f(u_i, u_i) = F(u_i)$$

• مساله ترافیک

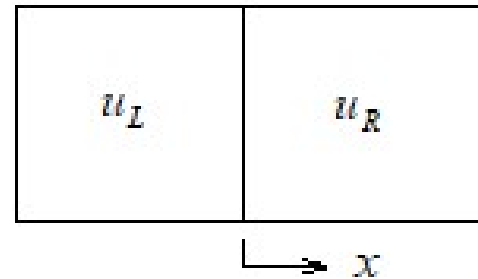


روش حجم محدود

• مساله ریمان

The Riemann problem:

$$u(x, 0) = \begin{cases} u_L, & x < 0 \\ u_R, & x > 0 \end{cases}$$



$$\frac{\partial u}{\partial t} + \frac{\partial F}{\partial x} = 0 \quad \& \quad F = \frac{u^2}{2} \rightarrow \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0$$

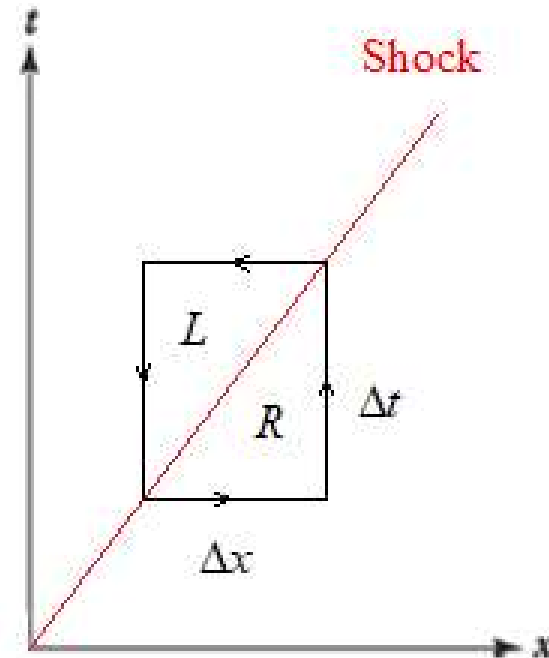
$$\text{Along } \frac{dx}{dt} = u : \quad \frac{du}{dt} = \frac{\partial u}{\partial t} + \frac{dx}{dt} \frac{\partial u}{\partial x} = 0 \quad \& \quad u = \text{const.}$$

Jump condition:

$$\oint_C u dx - F dt = 0$$

$$u_R \Delta x - u_L \Delta x - (F_R \Delta t - F_L \Delta t) = 0$$

$$\dot{x}_{\text{shock}} = \left. \frac{dx}{dt} \right|_{\text{shock}} = \lim_{\Delta x \text{ or } \Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{F_R - F_L}{u_R - u_L} = \frac{[F]}{[u]}$$

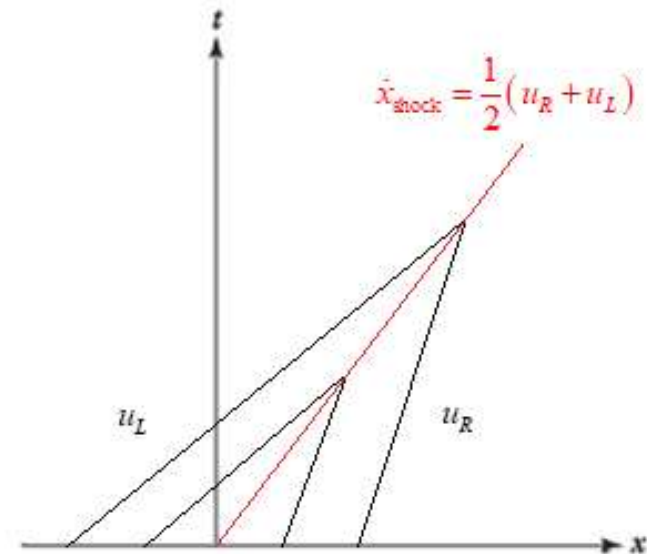


روش حجم محدود

Shock:

$$u(x, 0) = \begin{cases} u_L, & x < 0 \\ u_R, & x > 0 \end{cases} \quad \& \quad u_L > u_R$$

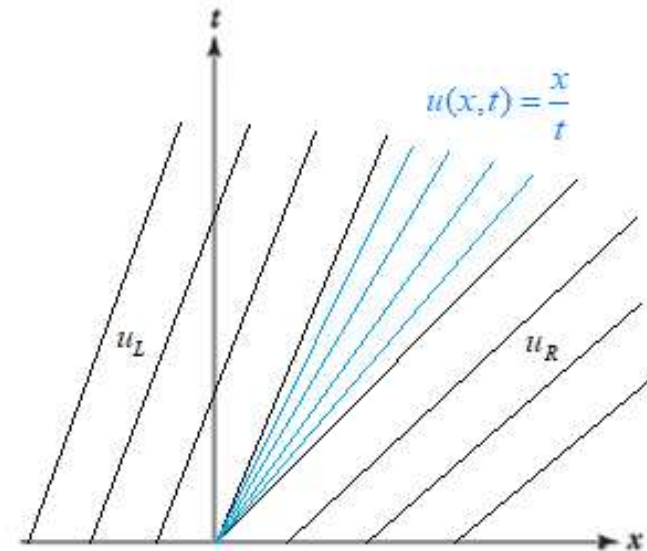
$$\dot{x}_{\text{shock}} = \frac{dx}{dt} \Big|_{\text{shock}} = \frac{F_R - F_L}{u_R - u_L} = \frac{1}{2} \frac{u_R^2 - u_L^2}{u_R - u_L} = \frac{1}{2} (u_R + u_L)$$



Fan:

$$u(x, 0) = \begin{cases} u_L, & x < 0 \\ u_R, & x > 0 \end{cases} \quad \& \quad u_L < u_R$$

$$u(x, t) = \begin{cases} u_L & x < u_L t \\ \frac{x}{t} & u_L t < x < u_R t \\ u_R & x > u_R t \end{cases}$$



روش حجم محدود

- روابط رانکین-هوگونیوت برای معادلات اوילر یک بعدی

One dimensional Euler eqs.:

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} = 0 \quad \& \quad Q = \begin{pmatrix} \rho \\ \rho u \\ E_t \end{pmatrix} \quad \& \quad F = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ (E_t + p)u \end{pmatrix}$$

$$\vec{\dot{x}}_{\text{shock}} = \frac{d\vec{x}}{dt}\bigg|_{\text{shock}} = \frac{F_R - F_L}{Q_R - Q_L}$$

For stationary shock:

$$F_R = F_L \quad \& \quad \begin{cases} \rho u|_L = \rho u|_R \\ \rho u^2 + p|_L = \rho u^2 + p|_R \\ (E_t + p)u|_L = (E_t + p)u|_R \end{cases} \quad \text{Rankine Hogoniout jump conditions}$$

روش حجم محدود

• حل گر ریمان تقریبی Roe

Roe's approximate Riemann solver:

$$\frac{\partial u}{\partial t} + \frac{\partial F}{\partial x} = 0 \quad \& \quad F = \frac{u^2}{2}$$

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + \frac{f_{i+\frac{1}{2}}^* - f_{i-\frac{1}{2}}^*}{\Delta x} = 0 \quad \& \quad f_{i\pm\frac{1}{2}}^* = \text{numerical flux}$$

- حل مساله خطی بجای مساله غیر خطی

$$\frac{\partial u}{\partial t} + \tilde{A} \frac{\partial u}{\partial x} = 0 \quad \& \quad \tilde{A} = \text{constant matrix dependent on local conditions}$$

- خواص ماتریس $\tilde{A}$ در روش Roe

$$1) \text{ For any } u_i, u_{i+1}: \quad F_{i+1} - F_i = \tilde{A}(u_{i+1} - u_i)$$

$$2) \text{ When } u = u_i, u_{i+1}, \text{ then: } \tilde{A}(u_i, u_{i+1}) = \tilde{A}(u, u) = \frac{\partial F}{\partial u} = u$$

3) $\tilde{A}$ has real eigenvalue(s) with linearly independent eigenvector(s):

$$\frac{\partial u}{\partial t} + \tilde{A} \frac{\partial u}{\partial x} = 0 \quad \& \quad \tilde{A} = \tilde{u} \rightarrow \tilde{u} = \tilde{u}_{i+\frac{1}{2}} = \frac{F_{i+1} - F_i}{u_{i+1} - u_i} = \begin{cases} \frac{1}{2}(u_i + u_{i+1}) & u_i \neq u_{i+1} \\ u_i & u_i = u_{i+1} \end{cases}$$

روش حجم محدود

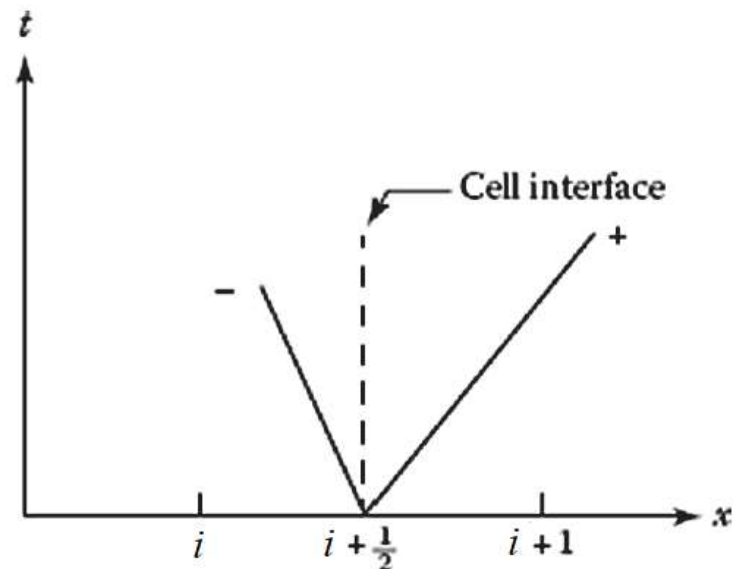
• حل گر ریمان تقریبی Roe

- تعیین شار برای روش Roe

$$\begin{cases} f_{i+\frac{1}{2}}^* = F_i + \delta F_{i+\frac{1}{2}}^- = F_i + \tilde{u}_{i+\frac{1}{2}}^- (u_{i+1} - u_i) & \& \delta F^- = \tilde{A}^- \delta u \\ f_{i+\frac{1}{2}}^* = F_{i+1} - \delta F_{i+\frac{1}{2}}^+ = F_{i+1} - \tilde{u}_{i+\frac{1}{2}}^+ (u_{i+1} - u_i) & \& \delta F^+ = \tilde{A}^+ \delta u \end{cases}$$

$$f_{i+\frac{1}{2}}^* = \frac{1}{2}(F_{i+1} + F_i) + \frac{1}{2}(\tilde{u}_{i+\frac{1}{2}}^- - \tilde{u}_{i+\frac{1}{2}}^+)(u_{i+1} - u_i)$$

$$\begin{cases} f_{i+\frac{1}{2}}^* = \frac{1}{2}(F_{i+1} + F_i) - \frac{1}{2}|\tilde{u}_{i+\frac{1}{2}}|(u_{i+1} - u_i) \\ f_{i-\frac{1}{2}}^* = \frac{1}{2}(F_{i-1} + F_i) - \frac{1}{2}|\tilde{u}_{i-\frac{1}{2}}|(u_{i+1} - u_i) \end{cases}$$



روش حجم محدود

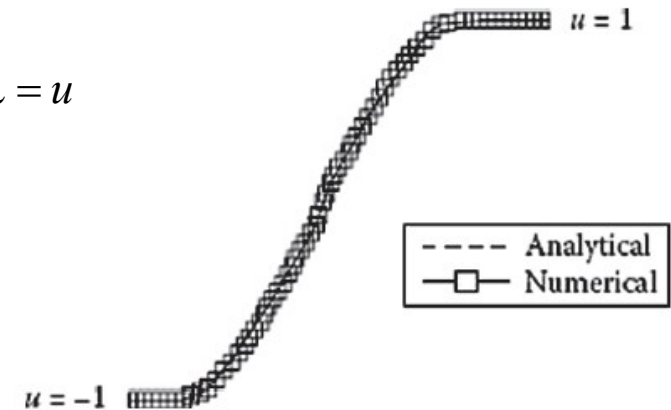
• حل گر ریمان تقریبی Roe

- اعتبار روش Roe در شوک تراکمی
- عدم اعتبار روش Roe در فن انبساطی
- عدم وجود اطلاعات یا (منحنی مشخصه) در نقطه صوتی / عدم وجود میرایی عددی در نقطه صوتی

Harten & Heyman (1983):

$$|\tilde{\lambda}|_{\max} = \begin{cases} |\tilde{\lambda}|_{i+\frac{1}{2}}, & |\tilde{\lambda}|_{i+\frac{1}{2}} \geq \varepsilon \\ \varepsilon, & |\tilde{\lambda}|_{i+\frac{1}{2}} < \varepsilon \end{cases} \quad \& \quad \tilde{\lambda}_{i+\frac{1}{2}} = \frac{1}{2}(\lambda_{i+1} + \lambda_i) \quad \& \quad \lambda = u$$

$$\varepsilon = \max \left[0, \left(\tilde{\lambda}_{i+\frac{1}{2}} - \lambda_i \right) \right] = \begin{cases} \frac{1}{2}(\lambda_{i+1} - \lambda_i) > 0 & \text{expansion fan} \\ 0 & \text{shock} \end{cases}$$



Expansion fan

Shock

--- Analytical
-□- Numerical

--- Analytical
-□- Numerical

$u = -1$

$u = 1$

$u = 1$

$u = 0$

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} = 0 \quad \& \quad Q$$

روش حجم محدود

One dimensional Euler eqs.:

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} = 0 \quad \& \quad Q = \begin{pmatrix} \rho \\ \rho u \\ E_t \end{pmatrix} \quad \& \quad F = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ (E_t + p)u \end{pmatrix}$$

$$\frac{\partial Q}{\partial t} + A \frac{\partial Q}{\partial x} = 0 \quad \& \quad A = \frac{\partial F}{\partial Q} = \begin{pmatrix} 0 & 1 & 0 \\ (\gamma-3)\frac{u^2}{2} & (3-\gamma)u & \gamma-1 \\ (\gamma-1)u^3 - \gamma\frac{E_t u}{\rho} & \gamma\frac{E_t}{\rho} - \frac{3}{2}(\gamma-1)u^2 & \gamma u \end{pmatrix}$$

Roe's scheme:

$$\frac{\partial Q}{\partial t} + \tilde{A} \frac{\partial Q}{\partial x} = 0 \quad \& \quad \tilde{A} = \text{Roe's average matrix}$$

$$1) F_{i+1} - F_i = \tilde{A}(Q_{i+1} - Q_i) \quad \& \quad \tilde{A} = \tilde{A}(Q_i, Q_{i+1})$$

$$2) \tilde{A}(Q_i, Q_i) = \tilde{A}(Q) = A(Q) = \frac{\partial F}{\partial Q}$$

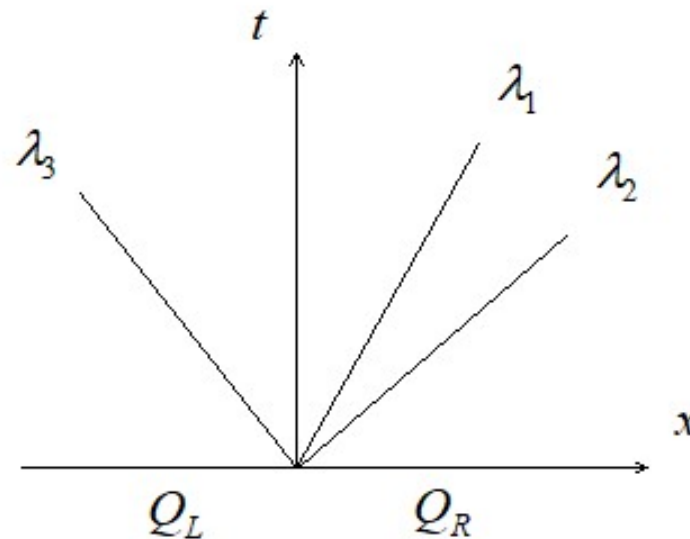
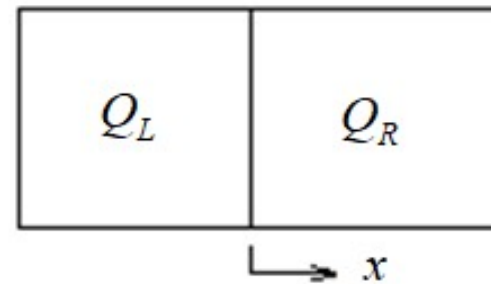
• روش Roe برای معادلات اویلر یک بعدی

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} = 0 \quad \& \quad Q$$

روش حجم محدود

• روش Roe برای معادلات اوایلر یک بعدی

3) $\tilde{A}$ has real eigenvalue(s) with linearly independent eigenvector(s)



$\hat{Q}_k = \text{const.}$ along λ_k

روش حجم محدود

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} = 0$$

$$\frac{Q_i^{n+1} - Q_i^n}{\Delta t} + \frac{f_{i+\frac{1}{2}}^* - f_{i-\frac{1}{2}}^*}{\Delta x} = 0 \quad \& \quad f_{i\pm\frac{1}{2}}^* = \text{numerical flux}$$

$$\begin{cases} f_{i+\frac{1}{2}}^* = F_i + \delta F_{i+\frac{1}{2}}^- & \& \delta F^- = \tilde{A}^- \delta Q \\ f_{i+\frac{1}{2}}^* = F_{i+1} - \delta F_{i+\frac{1}{2}}^+ & \& \delta F^+ = \tilde{A}^+ \delta Q \end{cases}$$

$$\delta Q = P^{-1} \delta \hat{Q} = (L^T M^{-1})^{-1} \delta \hat{Q} = M (L^T)^{-1} \delta \hat{Q} = R \delta \hat{Q} \quad \& \quad R = [R] = [R^{(1)}, R^{(2)}, R^{(3)}] = \begin{pmatrix} r_1^{(1)} & r_1^{(2)} & r_1^{(3)} \\ r_2^{(1)} & r_2^{(2)} & r_2^{(3)} \\ r_3^{(1)} & r_3^{(2)} & r_3^{(3)} \end{pmatrix}$$

$$\delta Q = \sum_j \delta \hat{Q}_j r^{(j)} = \delta \hat{Q}_1 \begin{pmatrix} 1 \\ u \\ u^2/2 \end{pmatrix} + \delta \hat{Q}_2 \begin{pmatrix} 1 \\ u+a \\ H+ua \end{pmatrix} \begin{pmatrix} \rho \\ 2a \end{pmatrix} + \delta \hat{Q}_3 \begin{pmatrix} 1 \\ u-a \\ H-ua \end{pmatrix} \begin{pmatrix} -\rho \\ 2a \end{pmatrix}$$

$$\delta \hat{Q}_1 = \delta \rho - \frac{1}{a^2} \delta p \quad \& \quad \delta \hat{Q}_2 = \delta u + \frac{1}{\rho a} \delta p \quad \& \quad \delta \hat{Q}_3 = \delta u - \frac{1}{\rho a} \delta p$$

$$\delta F = A \delta Q = A \sum_j \delta \hat{Q}_j r^{(j)} = \sum_j \lambda_j \delta \hat{Q}_j r^{(j)} \quad \& \quad \lambda_1 = u \quad \& \quad \lambda_2 = u + a \quad \& \quad \lambda_3 = u - a$$

• روش Roe برای معادلات اویلر یک بعدی

روش حجم محدود

- روش Roe برای معادلات اوایلر یک بعدی

$$\begin{cases} \delta F^- = A^- \delta Q = A^- \sum_j \delta \hat{Q}_j r^{(j)} = \sum_j \lambda_j^- \delta \hat{Q}_j r^{(j)} \\ \delta F^+ = A^+ \delta Q = A^+ \sum_j \delta \hat{Q}_j r^{(j)} = \sum_j \lambda_j^+ \delta \hat{Q}_j r^{(j)} \end{cases}$$

$$\begin{cases} f_{i+\frac{1}{2}}^* = F_i + \delta F_{i+\frac{1}{2}}^- = F_i + \sum_j \lambda_j^- \delta \hat{Q}_j r^{(j)} \\ f_{i+\frac{1}{2}}^* = F_{i+1} - \delta F_{i+\frac{1}{2}}^+ = F_{i+1} - \sum_j \lambda_j^+ \delta \hat{Q}_j r^{(j)} \end{cases} \quad \& \quad \begin{cases} \lambda_j^- = -|\lambda_j| \\ \lambda_j^+ = |\lambda_j| \end{cases}$$

$$f_{i+\frac{1}{2}}^* = \frac{1}{2} \left[(F_i + F_{i+1}) - \sum_j |\lambda_j| \delta \hat{Q}_j r^{(j)} \right] = \frac{1}{2} \left[(F_i + F_{i+1}) - |A| \sum_j \delta \hat{Q}_j r^{(j)} \right]$$

$$f_{i-\frac{1}{2}}^* = \frac{1}{2} \left[(F_i + F_{i-1}) - \sum_j |\lambda_j| \delta \hat{Q}_j r^{(j)} \right] = \frac{1}{2} \left[(F_i + F_{i-1}) - |A| \sum_j \delta \hat{Q}_j r^{(j)} \right]$$

- تعیین مقادیر ویژه و بردارهای ویژه با توجه ماتریس $\tilde{A}$ و برحسب مقادیر چپ و راست مرز مشترک

روش حجم محدود

- روش Roe برای معادلات اوایلر یک بعدی

- ایده استفاده از فرم مربع کامل (Quadratic function) در مساله ترافیک

$$\frac{\partial u}{\partial t} + \frac{\partial F}{\partial x} = 0 \quad \& \quad F = \frac{u^2}{2}$$

$$\frac{\partial u}{\partial t} + \tilde{A} \frac{\partial u}{\partial x} = 0 \quad \& \quad \tilde{u}_{i+\frac{1}{2}} = \frac{1}{2}(u_i + u_{i+1}) = \frac{1}{2}(u_L + u_R)$$

$$Z = \sqrt{\rho} \begin{pmatrix} 1 \\ u \\ H \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}$$

- بیان Q و F بر حسب Z (Q و F توابع مربعی از Z)

$$H = \frac{E_t + p}{\rho} \quad \& \quad E_t = \frac{p}{\gamma - 1} + \frac{1}{2} \rho u^2 \quad \& \quad p = \frac{\rho(\gamma - 1)}{\gamma} \left(H - \frac{1}{2} u^2 \right) \quad \& \quad E_t = \frac{\rho H}{\gamma} + \frac{1}{2} \frac{\rho(\gamma - 1)}{\gamma} u^2$$

$$Q = \begin{pmatrix} \rho \\ \rho u \\ E_t \end{pmatrix} = \begin{pmatrix} z_1^2 \\ z_1 z_2 \\ \frac{z_1 z_3}{\gamma} + \frac{\gamma - 1}{2\gamma} z_2^2 \end{pmatrix} \quad \& \quad F = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ (E_t + p)u \end{pmatrix} = \begin{pmatrix} z_1 z_2 \\ \frac{\gamma - 1}{\gamma} z_1 z_3 + \frac{1}{2} \frac{\gamma + 1}{\gamma} z_2^2 \\ z_2 z_3 \end{pmatrix}$$

روش حجم محدود

- روش Roe برای معادلات اوایلر یک بعدی

$$Q_{i+1} - Q_i = \tilde{B}(Z_{i+1} - Z_i) \quad \& \quad F_{i+1} - F_i = \tilde{C}(Z_{i+1} - Z_i)$$

$$\tilde{B} = \begin{pmatrix} 2\tilde{z}_1 & 0 & 0 \\ \tilde{z}_2 & \tilde{z}_1 & 0 \\ \frac{\tilde{z}_3}{\gamma} & \frac{\gamma-1}{\gamma}\tilde{z}_2 & \frac{\tilde{z}_1}{\gamma} \end{pmatrix} \quad \& \quad \tilde{C} = \begin{pmatrix} \tilde{z}_2 & \tilde{z}_1 & 0 \\ \frac{\gamma-1}{\gamma}\tilde{z}_3 & \frac{\gamma+1}{\gamma}\tilde{z}_2 & \frac{\gamma-1}{\gamma}\tilde{z}_1 \\ 0 & \tilde{z}_3 & \tilde{z}_2 \end{pmatrix}$$

$$F_{i+1} - F_i = \tilde{C} \tilde{B}^{-1} (Q_{i+1} - Q_i) = \tilde{A} (Q_{i+1} - Q_i)$$

$$\tilde{A} = \tilde{C} \tilde{B}^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ -\frac{(\gamma-3)\tilde{z}_2^2}{2\tilde{z}_1^2} & \frac{(\gamma-3)\tilde{z}_2^2}{\tilde{z}_1} & \gamma-1 \\ -\frac{\tilde{z}_2\tilde{z}_3}{\tilde{z}_1^2} + \frac{(\gamma-1)\tilde{z}_2^3}{2\tilde{z}_1^3} & -\frac{(\gamma-1)\tilde{z}_3^2}{2\tilde{z}_1^2} + \frac{\tilde{z}_3}{\tilde{z}_1} & \gamma \frac{\tilde{z}_2}{\tilde{z}_1} \end{pmatrix}$$

روش حجم محدود

• روش Roe برای معادلات اویلر یک بعدی

- محاسبه بردار Z در سطح تماس سلول برای ارضای شرط $A = \tilde{A}$

$$R_{i+\frac{1}{2}} = \sqrt{\frac{\rho_{i+1}}{\rho_i}}$$

$$\tilde{\rho}_{i+\frac{1}{2}} = \sqrt{\rho_i \rho_{i+1}} = R_{i+\frac{1}{2}} \rho_i$$

$$\tilde{u}_{i+\frac{1}{2}} = \frac{\tilde{z}_2}{\tilde{z}_1} = \frac{(u\sqrt{\rho})_{i+1} + (u\sqrt{\rho})_i}{\sqrt{\rho_{i+1}} + \sqrt{\rho_i}} = \frac{R_{i+\frac{1}{2}} u_{i+1} + u_i}{R_{i+\frac{1}{2}} + 1}$$

$$\tilde{H}_{i+\frac{1}{2}} = \frac{\tilde{z}_3}{\tilde{z}_1} = \frac{(H\sqrt{\rho})_{i+1} + (H\sqrt{\rho})_i}{\sqrt{\rho_{i+1}} + \sqrt{\rho_i}} = \frac{R_{i+\frac{1}{2}} H_{i+1} + H_i}{R_{i+\frac{1}{2}} + 1}$$

روش حجم محدود

• روش Roe برای معادلات اوילر یک بعدی

- مراحل روش Roe

Step 1: Calculate averaged values for at $i + \frac{1}{2}$:

$$\tilde{\rho}_{i+\frac{1}{2}} \quad \& \quad \tilde{u}_{i+\frac{1}{2}} \quad \& \quad \tilde{H}_{i+\frac{1}{2}} \quad \& \quad \tilde{a}^2 = (\gamma - 1) \left(\tilde{H} - \frac{1}{2} \tilde{u}^2 \right)$$

Step 2: Calculate the eigenvalues and eigenvectors of the linearized matrix $\tilde{A}(Q_i, Q_{i+1})$ at $i + \frac{1}{2}$:

$$\tilde{\lambda}_1 = \tilde{u} \quad \& \quad \tilde{\lambda}_2 = \tilde{u} + \tilde{a} \quad \& \quad \tilde{\lambda}_3 = \tilde{u} - \tilde{a}$$

$$\tilde{R}^{(1)} = \begin{pmatrix} \tilde{r}_1^{(1)} \\ \tilde{r}_2^{(1)} \\ \tilde{r}_3^{(1)} \end{pmatrix} = \begin{pmatrix} 1 \\ \tilde{u} \\ \tilde{u}^2 / 2 \end{pmatrix} \quad \& \quad \tilde{R}^{(2)} = \begin{pmatrix} \tilde{r}_1^{(2)} \\ \tilde{r}_2^{(2)} \\ \tilde{r}_3^{(2)} \end{pmatrix} = \begin{pmatrix} 1 \\ \tilde{u} + \tilde{a} \\ \tilde{H} + \tilde{u}\tilde{a} \end{pmatrix} \begin{pmatrix} \tilde{\rho} \\ 2\tilde{a} \end{pmatrix} \quad \& \quad \tilde{R}^{(3)} = \begin{pmatrix} \tilde{r}_1^{(3)} \\ \tilde{r}_2^{(3)} \\ \tilde{r}_3^{(3)} \end{pmatrix} = \begin{pmatrix} 1 \\ \tilde{u} - \tilde{a} \\ \tilde{H} - \tilde{u}\tilde{a} \end{pmatrix} \begin{pmatrix} -\tilde{\rho} \\ 2\tilde{a} \end{pmatrix}$$

Step 3: Calculate the wave amplitude at $i + \frac{1}{2}$:

$$\delta \hat{Q}_1 = \delta \rho - \frac{1}{\tilde{a}^2} \delta p \quad \& \quad \delta \hat{Q}_2 = \delta u + \frac{1}{\rho \tilde{a}} \delta p \quad \& \quad \delta \hat{Q}_3 = \delta u - \frac{1}{\tilde{\rho} \tilde{a}} \delta p$$

$$\delta \rho_{i+\frac{1}{2}} = \rho_{i+1} - \rho_i \quad \& \quad \delta p_{i+\frac{1}{2}} = p_{i+1} - p_i \quad \& \quad \delta u_{i+\frac{1}{2}} = u_{i+1} - u_i$$

روش حجم محدود

• روش Roe برای معادلات اویلر یک بعدی

- مراحل روش Roe

Step 4: Calculate the numerical flux :

$$f_{i+\frac{1}{2}}^* = \frac{1}{2} \left[(F_i + F_{i+1}) - \sum_j |\tilde{\lambda}_j| \delta \hat{Q}_j \tilde{r}^{(j)} \right] = \frac{1}{2} \left[(F_i + F_{i+1}) - |\tilde{A}| \sum_j \delta \hat{Q}_j \tilde{r}^{(j)} \right] = \frac{1}{2} \left[(F_i + F_{i+1}) - |\tilde{A}| \delta Q \right]$$

$$f_{i-\frac{1}{2}}^* = \frac{1}{2} \left[(F_i + F_{i-1}) - \sum_j |\tilde{\lambda}_j| \delta \hat{Q}_j \tilde{r}^{(j)} \right] = \frac{1}{2} \left[(F_i + F_{i-1}) - |\tilde{A}| \sum_j \delta \hat{Q}_j \tilde{r}^{(j)} \right] = \frac{1}{2} \left[(F_i + F_{i-1}) - |\tilde{A}| \delta Q \right]$$

$$\delta Q_{i+\frac{1}{2}} = Q_{i+1} - Q_i \quad \& \quad \delta Q_{i-\frac{1}{2}} = Q_i - Q_{i-1}$$

روش حجم محدود

• روش Roe برای حل معادلات اویلر دوبعدی

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0 \quad \& \quad Q = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ E_t \end{pmatrix} \quad \& \quad F = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (E_t + p)u \end{pmatrix} \quad \& \quad G = \begin{pmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ (E_t + p)v \end{pmatrix}$$

$$\iint_A \frac{\partial Q}{\partial t} dA + \iint_A \left(\frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} \right) dA = 0$$

$$\frac{\partial}{\partial t} \iint_A Q dA + \iint_A \vec{\nabla} \cdot \vec{H} dA = 0 \quad \& \quad \vec{H} = F \hat{i} + G \hat{j}$$

$$\frac{\partial}{\partial t} \iint_A Q dA + \oint_C \vec{H} \cdot \hat{n} dl = 0 \quad \& \quad \vec{H} \cdot \hat{n} = F \hat{n}_x + G \hat{n}_y = \vec{V} \cdot \hat{n} \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ (E_t + p) \end{pmatrix} + p \begin{pmatrix} 0 \\ \hat{n}_x \\ \hat{n}_y \\ 0 \end{pmatrix}$$

روش حجم محدود

• روش Roe برای حل معادلات اوایلر دوبعدی

$$\frac{d}{dt}(Q_{i,j} A_{i,j}) + R(Q_{i,j}) = 0 \quad \& \quad R(Q_{i,j}) = \sum_{m=1}^{3 \text{ or } 4} \vec{H} \cdot d\vec{l} = \sum_{m=1}^{3 \text{ or } 4} (H_n)_m dl_m$$

$$H_m = \frac{1}{2} [H(Q_L) + H(Q_R) - |\tilde{A}|(Q_R - Q_L)] \quad \& \quad \tilde{A} = \frac{\partial H}{\partial Q}$$

$$|\tilde{A}|(Q_R - Q_L) = |\Delta \tilde{H}_1| + |\Delta \tilde{H}_2| + |\Delta \tilde{H}_3|$$

$$|\Delta \tilde{H}_1| = \left| \vec{\tilde{V}} \cdot \hat{n} \right| \left(\Delta \rho - \frac{1}{\tilde{a}^2} \Delta p \right) \begin{pmatrix} 1 \\ \tilde{u} \\ \tilde{v} \\ \frac{1}{2}(\tilde{u}^2 + \tilde{v}^2) \end{pmatrix} + \tilde{\rho} \begin{pmatrix} 0 \\ \Delta u - \hat{n}_x \Delta \vec{V} \cdot \hat{n} \\ \Delta v - \hat{n}_y \Delta \vec{V} \cdot \hat{n} \\ \tilde{u} \Delta u + \tilde{v} \Delta v - \vec{V} \cdot \hat{n} \Delta \vec{V} \cdot \hat{n} \end{pmatrix}$$

$$|\Delta \tilde{H}_{2,3}| = \left| \vec{\tilde{V}} \cdot \hat{n} \pm \tilde{a} \right| \left(\frac{\Delta p \pm \tilde{\rho} \tilde{a} \Delta \vec{V} \cdot \hat{n}}{2\tilde{a}^2} \right) \begin{pmatrix} 1 \\ \tilde{u} \pm \hat{n}_x \tilde{a} \\ \tilde{v} \pm \hat{n}_y \tilde{a} \\ \tilde{H} \pm \vec{V} \cdot \hat{n} \tilde{a} \end{pmatrix}$$

روش حجم محدود

• روش Roe برای حل معادلات اوایلر دوبعدی

$$\tilde{\rho} = \sqrt{\rho_L \rho_R} \quad \& \quad \tilde{u} = \frac{u_L + u_R \sqrt{\rho_R / \rho_L}}{1 + \sqrt{\rho_R / \rho_L}} \quad \& \quad \tilde{v} = \frac{v_L + v_R \sqrt{\rho_R / \rho_L}}{1 + \sqrt{\rho_R / \rho_L}} \quad \& \quad \tilde{H} = \frac{H_L + H_R \sqrt{\rho_R / \rho_L}}{1 + \sqrt{\rho_R / \rho_L}}$$

$$\tilde{a}^2 = (\gamma - 1) \left(\tilde{H} - \frac{1}{2} (\tilde{u}^2 + \tilde{v}^2) \right)$$

$$H(Q_R) - H(Q_L) = |\tilde{A}| (Q_R - Q_L)$$

$$\Delta t_{i,j} \leq CFL \frac{A_{i,j}}{\sum_{m=1}^N (\vec{V} \cdot \hat{n} + a)_m \Delta l_m / N}$$

روش حجم محدود

- روش آپویند دقت مرتبه بالا

- بیان تغییرات تابع u در سلول i ام برحسب موقعیت و مقدار متوسط u در آن سلول

$$u(x) = u_i + \frac{1}{\Delta x}(x - x_i)\delta_i u + \frac{3k}{2(\Delta x)^2} \left[(x - x_i)^2 - \frac{(\Delta x)^2}{12} \right] \delta_i^2 u \quad \& \quad u_i = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} u(x) dx \quad \& \quad x_{i-\frac{1}{2}} < x < x_{i+\frac{1}{2}}$$

- سری تیلور با دقت مرتبه ۳ به ازاء $(3k = 1)$

$$\begin{cases} x = x_i + \frac{\Delta x}{2} \\ x = x_i - \frac{\Delta x}{2} \end{cases} \rightarrow \begin{cases} u_{i+\frac{1}{2}}^L = u_i + \frac{1}{2}\delta_i u + \frac{k}{4}\delta_i^2 u \\ u_{i-\frac{1}{2}}^R = u_i - \frac{1}{2}\delta_i u + \frac{k}{4}\delta_i^2 u \end{cases} \quad \& \quad \begin{cases} \delta_i u = \frac{1}{2}(u_{i+1} - u_{i-1}) \\ \delta_i^2 u = u_{i+1} - 2u_i + u_{i-1} \end{cases}$$

$$\begin{cases} u_{i+\frac{1}{2}}^L = u_i + \frac{1}{4}(1-k)(u_i - u_{i-1}) + \frac{1}{4}(1+k)(u_{i+1} - u_i) \\ u_{i-\frac{1}{2}}^R = u_i - \frac{1}{4}(1+k)(u_i - u_{i-1}) - \frac{1}{4}(1-k)(u_{i+1} - u_i) \end{cases}$$

- برونیابی بردار حالت در سطوح حجم کنترل

روش حجم محدود

- روش آپویند دقت مرتبه بالا

Montone Upstream-centered Schemes for Conservation Laws (MUSCL), van Leer (1979):

$$\begin{cases} u_{i+\frac{1}{2}}^L = u_i + \frac{\varepsilon}{4} \left[(1-k)(u_i - u_{i-1}) + \frac{1}{4}(1+k)(u_{i+1} - u_i) \right] \\ u_{i+\frac{1}{2}}^R = u_{i+1} - \frac{\varepsilon}{4} \left[(1+k)(u_{i+1} - u_i) + \frac{1}{4}(1-k)(u_{i+2} - u_{i+1}) \right] \end{cases} \quad \& \quad \varepsilon = \begin{cases} 0 & \text{1st-order scheme} \\ 1 & \text{high-order scheme} \end{cases}$$

$$\begin{aligned} u_{i+\frac{1}{2}}^R - u_{i+\frac{1}{2}}^L &= -\frac{1}{4}(1-k)(u_{i+2} - 3u_{i+1} + 3u_i - u_{i-1}) \\ &\approx -\frac{1}{4}(1-k)(\Delta x)^3 \frac{\partial^3 u}{\partial x^3} + O(\Delta x)^4 \end{aligned} \quad \begin{cases} \text{3rd-order accuracy } k \neq 1 \\ \text{4th-order accuracy } k = 1 \end{cases}$$

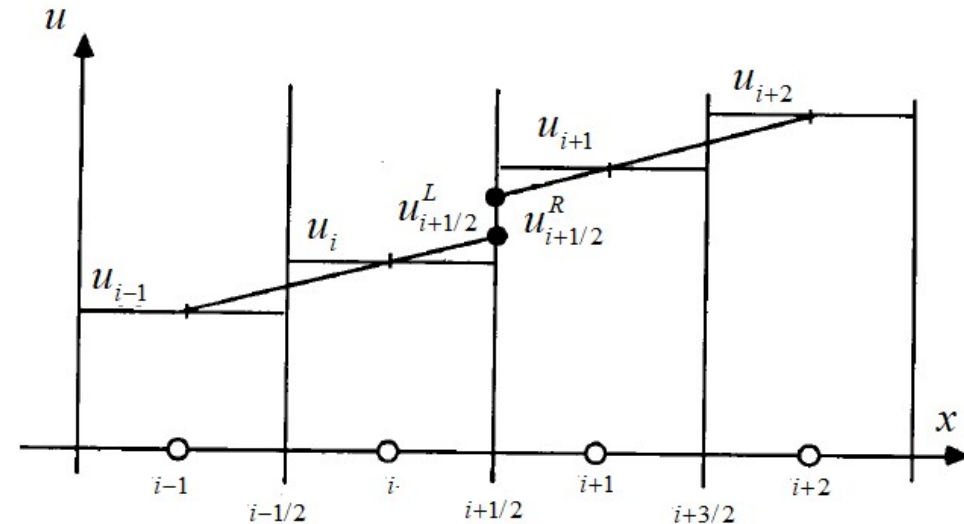
روش حجم محدود

- روش آپویند دقت مرتبه بالا

$$\begin{cases} u_{i+\frac{1}{2}}^L = u_i + \frac{\varepsilon}{4} \left[(1-k)(u_i - u_{i-1}) + \frac{1}{4}(1+k)(u_{i+1} - u_i) \right] \\ u_{i+\frac{1}{2}}^R = u_{i+1} - \frac{\varepsilon}{4} \left[(1+k)(u_{i+1} - u_i) + \frac{1}{4}(1-k)(u_{i+2} - u_{i+1}) \right] \end{cases}$$

2nd-order fully one-sided scheme ($k = -1$):

$$\begin{cases} u_{i+\frac{1}{2}}^L = u_i + \frac{\varepsilon}{2}(u_i - u_{i-1}) & \& \delta u_{i-\frac{1}{2}} = u_i - u_{i-1} \\ u_{i+\frac{1}{2}}^R = u_{i+1} - \frac{\varepsilon}{2}(u_{i+2} - u_{i+1}) & \& \delta u_{i+\frac{3}{2}} = u_{i+2} - u_{i+1} \end{cases}$$



روش حجم محدود

- روش آپویند دقت مرتبه بالا

$$\begin{cases} u_{i+\frac{1}{2}}^L = u_i + \frac{\varepsilon}{4} \left[(1-k)(u_i - u_{i-1}) + \frac{1}{4}(1+k)(u_{i+1} - u_i) \right] \\ u_{i+\frac{1}{2}}^R = u_{i+1} - \frac{\varepsilon}{4} \left[(1+k)(u_{i+1} - u_i) + \frac{1}{4}(1-k)(u_{i+2} - u_{i+1}) \right] \end{cases}$$

2nd-order central scheme ($k=1$):

$$\begin{cases} u_{i+\frac{1}{2}}^L = u_i + \frac{\varepsilon}{2}(u_{i+1} - u_i) \\ u_{i+\frac{1}{2}}^R = u_{i+1} - \frac{\varepsilon}{2}(u_{i+1} - u_i) \end{cases} \quad \text{for } \varepsilon = 1 \quad \rightarrow \quad \begin{cases} u_{i+\frac{1}{2}}^L = \frac{1}{2}(u_{i+1} + u_i) \\ u_{i+\frac{1}{2}}^R = \frac{1}{2}(u_{i+1} + u_i) \end{cases} \quad \text{Averaging}$$

روش حجم محدود

3rd-order scheme ($k = 1/3$):

$$\begin{cases} u_{i+\frac{1}{2}}^L = u_i + \frac{\varepsilon}{4} \left[(1-k)(u_i - u_{i-1}) + \frac{1}{4}(1+k)(u_{i+1} - u_i) \right] \\ u_{i+\frac{1}{2}}^R = u_{i+1} - \frac{\varepsilon}{4} \left[(1+k)(u_{i+1} - u_i) + \frac{1}{4}(1-k)(u_{i+2} - u_{i+1}) \right] \end{cases}$$

• روش آپویند دقت مرتبه بالا

- استفاده از محدود کننده‌ها (Limiter)

$$\begin{cases} u_{i+\frac{1}{2}}^L = u_i + \frac{\varepsilon}{4} \left[(1-k)(u_i - u_{i-1})\psi_{i-\frac{1}{2}}^+ + \frac{1}{4}(1+k)(u_{i+1} - u_i)\psi_{i+\frac{1}{2}}^- \right] \\ u_{i+\frac{1}{2}}^R = u_{i+1} - \frac{\varepsilon}{4} \left[(1+k)(u_{i+1} - u_i)\psi_{i+\frac{3}{2}}^- + \frac{1}{4}(1-k)(u_{i+2} - u_{i+1})\psi_{i+\frac{1}{2}}^+ \right] \end{cases}$$

$$\begin{cases} \psi_{i-\frac{1}{2}}^+ = \psi_{i-\frac{1}{2}}^+ \left(r_{i-\frac{1}{2}}^+ \right) & \& \quad r^L = r_{i-\frac{1}{2}}^+ = \frac{\delta u_{i+\frac{1}{2}}}{\delta u_{i-\frac{1}{2}}} = \frac{u_{i+1} - u_i}{u_i - u_{i-1}} \\ \psi_{i+\frac{1}{2}}^- = \psi_{i-\frac{1}{2}}^+ \left(r_{i+\frac{1}{2}}^- \right) & \& \quad \frac{1}{r^L} = r_{i+\frac{1}{2}}^- = \frac{\delta u_{i-\frac{1}{2}}}{\delta u_{i+\frac{1}{2}}} = \frac{u_i - u_{i-1}}{u_{i+1} - u_i} \end{cases}$$

روش حجم محدود

- روش آپویند دقت مرتبه بالا

$$u_{i+\frac{1}{2}}^L = u_i + \frac{\varepsilon}{4} \left[(1-k)(u_i - u_{i-1})\psi_{i-\frac{1}{2}}^+ + (1+k)(u_{i+1} - u_i)\psi_{i+\frac{1}{2}}^- \right]$$

$$u_{i+\frac{1}{2}}^L = u_i + \frac{\varepsilon}{4} \left[(1-k)\psi_{i-\frac{1}{2}}^+ + (1+k)\frac{\delta u_{i+\frac{1}{2}}}{\delta u_{i-\frac{1}{2}}} \psi_{i+\frac{1}{2}}^- \right] \delta u_{i-\frac{1}{2}}$$

$$\begin{cases} u_{i+\frac{1}{2}}^L = u_i + \frac{\varepsilon}{4} \left[(1-k)\psi(r^L) + (1+k)r^L \psi\left(\frac{1}{r^L}\right) \right] \delta u_{i-\frac{1}{2}} \\ u_{i+\frac{1}{2}}^R = u_{i+1} - \frac{\varepsilon}{4} \left[(1+k)r^R \psi\left(\frac{1}{r^R}\right) + (1-k)\psi(r^R) \right] \delta u_{i+\frac{3}{2}} \end{cases}$$

$$\begin{cases} r_{i+\frac{3}{2}}^- = \frac{\delta u_{i+\frac{1}{2}}}{\delta u_{i+\frac{3}{2}}} = \frac{u_{i+1} - u_i}{u_{i+2} - u_{i+1}} = r^R \\ r_{i+\frac{1}{2}}^+ = \frac{\delta u_{i+\frac{3}{2}}}{\delta u_{i+\frac{1}{2}}} = \frac{u_{i+2} - u_{i+1}}{u_{i+1} - u_i} = \frac{1}{r^R} \end{cases}$$

روش حجم محدود

- پیش شرطی سازی معادلات اوایلر تراکم پذیر دوبعدی

Preconditioning of 2D compressible Euler eqs.:

$$\frac{\partial Q_c}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0 \quad \& \quad Q_c = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ E_t \end{pmatrix} \quad \& \quad F = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (E_t + p)u \end{pmatrix} \quad \& \quad G = \begin{pmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ (E_t + p)v \end{pmatrix}$$

$$\frac{\partial Q_c}{\partial t} + \frac{\partial F}{\partial Q_c} \frac{\partial Q_c}{\partial x} + \frac{\partial G}{\partial Q_c} \frac{\partial Q_c}{\partial y} = 0$$

$$\frac{\partial Q_c}{\partial t} + A_c \frac{\partial Q_c}{\partial x} + B_c \frac{\partial Q_c}{\partial y} = 0 \quad \& \quad A_c = \frac{\partial F}{\partial Q_c} \quad \& \quad B_c = \frac{\partial G}{\partial Q_c}$$

Eigenvalues of the Jacobian matrix $A_c = \frac{\partial F}{\partial Q_c}$: $\lambda = u, u, u + a, u - a$

$$\text{At low speeds: } u \rightarrow 0 \quad \rightarrow \quad \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{u + a}{u} \rightarrow \infty \quad \& \quad \Delta t \leq CFL \frac{\Delta x_{\min}}{\lambda_{\max}}$$

روش حجم محدود

- پیش شرطی سازی معادلات اوایلر تراکم پذیر دوبعدی

$$\frac{\partial Q_c}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0$$

$$\frac{\partial Q_c}{\partial Q_p} \frac{\partial Q_p}{\partial t} + \frac{\partial F}{\partial Q_p} \frac{\partial Q_p}{\partial x} + \frac{\partial G}{\partial Q_p} \frac{\partial Q_p}{\partial y} = 0 \quad \& \quad \Gamma = \frac{\partial Q_c}{\partial Q_p}$$

$$\frac{\partial Q_p}{\partial t} + \Gamma^{-1} A_p \frac{\partial Q_p}{\partial x} + \Gamma^{-1} B_p \frac{\partial Q_p}{\partial y} = 0$$

$$Q_c = (\rho, \rho u, \rho v, E_t)^T \quad \& \quad Q_p = (p, u, v, T)^T$$

$$\Gamma = \begin{pmatrix} \rho_p & 0 & 0 & \rho_T \\ \rho_p u & 0 & 0 & \rho_T u \\ \rho_p v & 0 & 0 & \rho_T v \\ \rho_p H - 1 & \rho u & \rho v & \rho_T H + \rho c_p \end{pmatrix} \quad \& \quad \Gamma = \begin{pmatrix} \rho_p & 0 & 0 & 0 \\ \rho_p u & 0 & 0 & 0 \\ \rho_p v & 0 & 0 & 0 \\ \rho_p H - 1 & \rho u & \rho v & \rho c_p \end{pmatrix}$$

$$\rho = \rho(p, T)$$

$$\rho = \rho(p)$$

روش حجم محدود

- پیش شرطی سازی معادلات اوایلر تراکم پذیر دوبعدی

$$a^2 = p_\rho = \frac{1}{\rho_p} = \frac{\partial p}{\partial \rho} = \text{Acoustic wave speed} \quad \rightarrow \quad \beta^2 = \text{artificial acoustic wave speed}$$

$$\Gamma = \begin{pmatrix} \frac{1}{\beta^2} & 0 & 0 & 0 \\ \frac{1}{\beta^2} u & 0 & 0 & 0 \\ \frac{1}{\beta^2} v & 0 & 0 & 0 \\ \frac{1}{\beta^2} H - 1 & \rho u & \rho v & \rho c_p \end{pmatrix}$$

Choi-Merkel preconditioner:

$$\beta^2 = \begin{cases} u_r^2 & u^2 < u_r^2 \\ u^2 & u_r^2 < u^2 < a^2 \\ a^2 & u^2 > a^2 \end{cases} \quad \& \quad \bar{\beta}^2 = \frac{\beta^2}{a^2} = \begin{cases} M_r^2 & M^2 < M_r^2 \\ M^2 & M_r^2 < M^2 < 1 \\ 1 & M^2 > 1 \end{cases} \quad \& \quad M_r = 10^{-5} - 10^{-3}$$

روش حجم محدود

- پیش شرطی سازی معادلات اوایلر تراکم پذیر دوبعدی

Eigenvalues of the Jacobian matrix $\tilde{A} = \Gamma(A_p n_x + B_p n_y)$

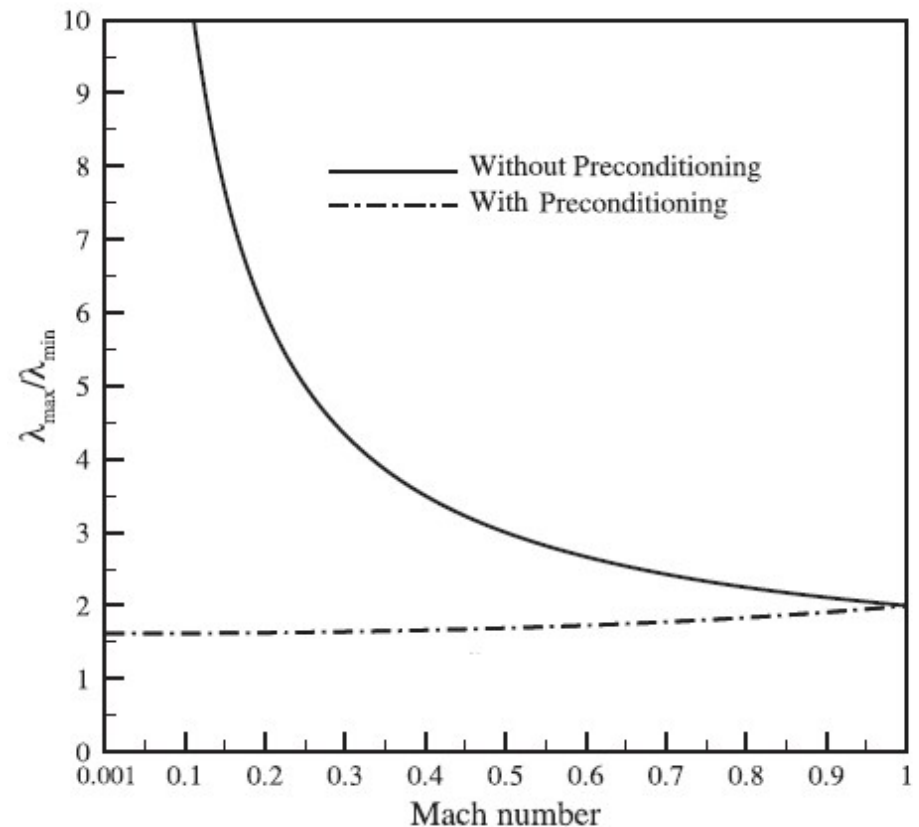
$$\lambda = \tilde{U}_T, \tilde{U}_T, \tilde{U}_T + \tilde{a}, \tilde{U}_T - \tilde{a}$$

$$\tilde{U}_T = U_T \frac{1 + \bar{\beta}^2}{2}$$

$$\tilde{a} = \frac{1}{2} \sqrt{U_T (1 - \bar{\beta}^2) + 4\beta^2}$$

$$U_T = un_x + vn_y$$

$$\bar{\beta}^2 = \frac{\beta^2}{a^2} = 1 \quad \rightarrow \quad \lambda = U_T, U_T, U_T + a, U_T - a$$



روش حجم محدود

- انتگرال گیری ضمنی دوزمانه

Implicit dual-time stepping method :

$$\frac{d}{dt}(Q_{i,j}^{n+1} A_{i,j}^{n+1}) + (H_i)_{i,j}^{n+1} - (H_v)_{i,j}^{n+1} - D_{i,j}^{n+1} = 0$$

$$\frac{3Q_{i,j}^{n+1} A_{i,j}^{n+1} - 4Q_{i,j}^n A_{i,j}^n + Q_{i,j}^{n-1} A_{i,j}^{n-1}}{2\Delta t} + (H_i)_{i,j}^{n+1} - (H_v)_{i,j}^{n+1} - D_{i,j}^{n+1} = 0$$

For nonmoving grid : $A_{i,j}^{n+1} = A_{i,j}^n = A_{i,j}^{n-1}$

$$\frac{3Q_{i,j}^{n+1} - 4Q_{i,j}^n + Q_{i,j}^{n-1}}{2\Delta t} + \frac{1}{A_{i,j}} \left\{ (H_i)_{i,j}^{n+1} - (H_v)_{i,j}^{n+1} - D_{i,j}^{n+1} \right\} = 0$$

Using the pseudo-time τ :

$$\frac{dQ_{i,j}^{n+1}}{d\tau} + R^*(Q_{i,j}^{n+1}) = 0 \quad \& \quad R^*(Q_{i,j}^{n+1}) = \frac{3Q_{i,j}^{n+1} - 4Q_{i,j}^n + Q_{i,j}^{n-1}}{2\Delta t} + \frac{1}{A_{i,j}} \left\{ (H_i)_{i,j}^{n+1} - (H_v)_{i,j}^{n+1} - D_{i,j}^{n+1} \right\}$$

$$\frac{dQ_{i,j}^{n+1}}{d\tau} \rightarrow 0 \quad \rightarrow \quad R^*(Q_{i,j}^{n+1}) = \frac{3Q_{i,j}^{n+1} - 4Q_{i,j}^n + Q_{i,j}^{n-1}}{2\Delta t} + \frac{1}{A_{i,j}} \left\{ (H_i)_{i,j}^{n+1} - (H_v)_{i,j}^{n+1} - D_{i,j}^{n+1} \right\} \rightarrow 0$$

روش حجم محدود

- انتگرال گیری ضمنی دوزمانه

$$\frac{dQ_{i,j}^{n+1}}{d\tau} + R^*(Q_{i,j}^{n+1}) = 0 \quad \& \quad R^*(Q_{i,j}^{n+1}) = \frac{3Q_{i,j}^{n+1} - 4Q_{i,j}^n + Q_{i,j}^{n-1}}{2\Delta t} + \frac{1}{A_{i,j}} \left\{ (H_i)_{i,j}^{n+1} - (H_v)_{i,j}^{n+1} - D_{i,j}^{n+1} \right\}$$

$$Q^{(0)} = (Q^{n+1})^m$$

$$Q^{(1)} = Q^{(0)} + \alpha_1 \Delta t R^*(Q^{(0)})$$

$$Q^{(2)} = Q^{(0)} + \alpha_2 \Delta t R^*(Q^{(1)})$$

⋮

$$Q^{(k-1)} = Q^{(0)} + \alpha_{k-1} \Delta t R^*(Q^{(k-2)})$$

$$Q^{(k)} = Q^{(0)} + \alpha_k \Delta t R^*(Q^{(k-1)})$$

$$(Q^{n+1})^{m+1} = Q^{(k)}$$

$$R^*(Q_{i,j}^{(k)}) = \frac{3Q_{i,j}^k - 4Q_{i,j}^n + Q_{i,j}^{n-1}}{2\Delta t} + \frac{1}{A_{i,j}} \left\{ (H_i)_{i,j}^k - (H_v)_{i,j}^k - D_{i,j}^0 \right\}$$

موفق باشید.